

If all of the settings were properly selected, EPLAN can now generate the assignment list. Use the **EXTRAS** button in the **ADDRESSES/ASSIGNMENT LISTS** dialog to select **EXPORT ASSIGNMENT LISTS**. In the **EXPORT ASSIGNMENT LISTS** dialog, you can check the settings once more and adjust the settings if necessary. Depending on the existing CPUs (there might be several controls in the project), select the correct option before an export in the **ADDRESSES/ASSIGNMENT LISTS** dialog.

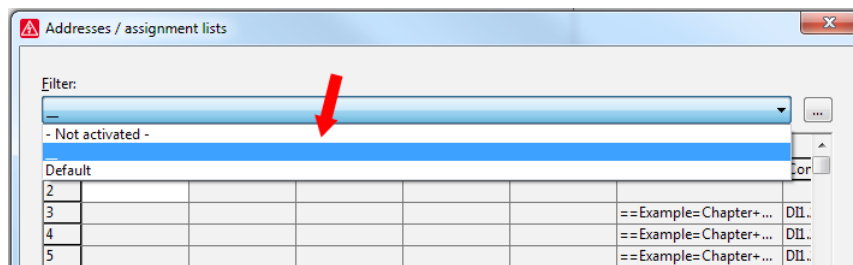


Fig. 5.152 CPU filter

You then simply confirm this dialog with **OK**. EPLAN generates the assignment list in the desired PLC software format and saves it in the specified directory.

5.6.5 Address

EPLAN allows simple addressing or readdressing of PLC terminals via the **ADDRESS** function in the **PROJECT DATA/PLC** menu. The functions in the logical PLC overview and the functions in the schematic for the PLC terminals in distributed view will be re-addressed or addressed for the first time.



NOTE: In order for all PLC functions to be addressed, it is sufficient to select the PLC card in the PLC navigator (but it should no longer have any messages from message management, that is, it must be “error-free”). This ensures that all of the functions, those of the logical PLC overview and the PLC functions represented in distributed view, are included and subsequently addressed.

In the PLC navigator, you select the card to be addressed. Then you call up the **ADDRESS** function in the **PROJECT DATA/PLC** menu.



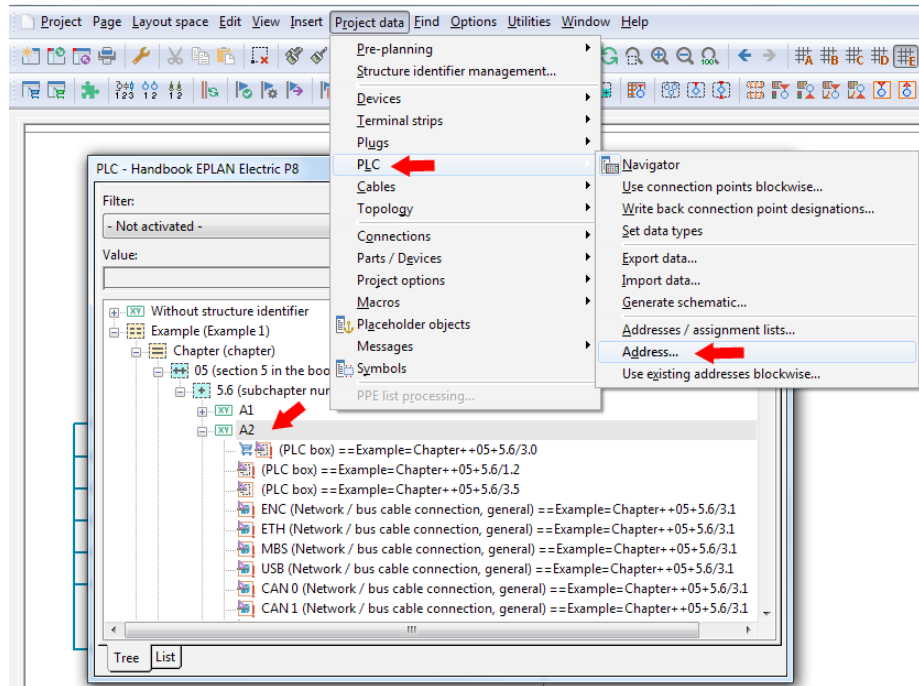


Fig. 5.153 Selecting the card to be addressed and calling up addressing

EPLAN now opens the **READDRESS PLC CONNECTION POINTS** dialog.

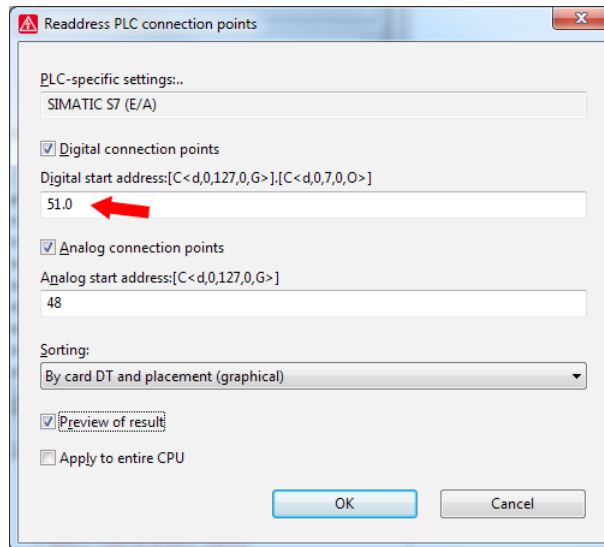
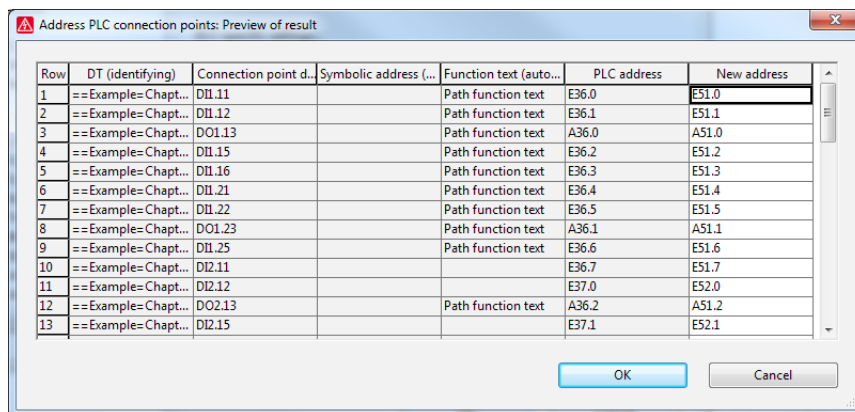


Fig. 5.154 Defining start addresses

In the **Digital start address** and **Analog start address** fields, the new address is entered *without* a prefix. EPLAN obtains the relevant prefix, meaning whether an entry begins with E or with I, according to the selected scheme, which is entered in the *PLC-specific settings* selection field. It is possible, however, to enter a prefix in the start address entry field, an “IX” for example. EPLAN would then use this primarily during addressing, and ignore the setting of the *PLC-specific settings* scheme.

I recommend leaving the *Preview of result* parameter switched on. When you click **OK**, the **ADDRESS PLC CONNECTION POINTS/PREVIEW OF RESULT** dialog is displayed, and you can use the **CANCEL** button to stop the process if necessary.



Row	DT (identifying)	Connection point d...	Symbolic address (...)	Function text (auto...	PLC address	New address
1	==Example= Chapt...	D11.11		Path function text	E36.0	E51.0
2	==Example= Chapt...	D11.12		Path function text	E36.1	E51.1
3	==Example= Chapt...	D01.13		Path function text	A36.0	A51.0
4	==Example= Chapt...	D11.15		Path function text	E36.2	E51.2
5	==Example= Chapt...	D11.16		Path function text	E36.3	E51.3
6	==Example= Chapt...	D11.21		Path function text	E36.4	E51.4
7	==Example= Chapt...	D11.22		Path function text	E36.5	E51.5
8	==Example= Chapt...	D01.23		Path function text	A36.1	A51.1
9	==Example= Chapt...	D11.25		Path function text	E36.6	E51.6
10	==Example= Chapt...	D12.11			E36.7	E51.7
11	==Example= Chapt...	D12.12			E37.0	E52.0
12	==Example= Chapt...	D02.13		Path function text	A36.2	A51.2
13	==Example= Chapt...	D12.15			E37.1	E52.1

Address PLC connection points

Fig. 5.155 Checking readdressing

If the result is correct, you click **OK** to confirm the **ADDRESS CONNECTION POINTS/PREVIEW OF RESULT** dialog and EPLAN readdresses the PLC functions of the selected cards in the entire project.

5.6.6 New

As in the other navigators, new devices with a function definition can also be created in the **PLC navigator**. The **NEW...** item in the popup menu is used for this.

EPLAN opens the **FUNCTION DEFINITIONS** dialog. In the PLC navigator you can only select PLC function definitions. After applying the function definition by clicking the **OK** button, EPLAN opens the **PROPERTIES (COMPONENTS) PLC BOX** dialog. Here you must make the desired entries and save the properties. You can also immediately perform a *device selection* for the new PLC card.

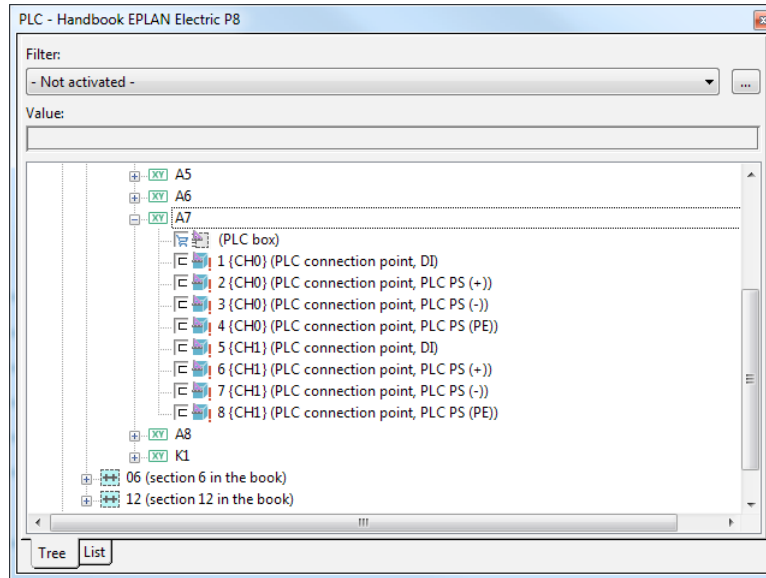


Fig. 5.156 New, unplaced PLC card in the navigator

The new device is stored in the **PLC navigator**, with a function definition of **PLC BOX** and the selected part's function definitions as an unplaced PLC box.

5.6.7 New functions

To add additional functions to devices in the PLC navigator, you use the **NEW FUNCTIONS...** function. This is started in the usual way via the popup menu in the **PLC navigator**.



EPLAN opens the **GENERATE FUNCTIONS** dialog. In the **Function definition** field, you can select a *PLC connection point, DI* (for example) via the  button and the subsequent **FUNCTION DEFINITIONS** dialog. For a card with 20 connection points, the numbering pattern is set to 1–20.

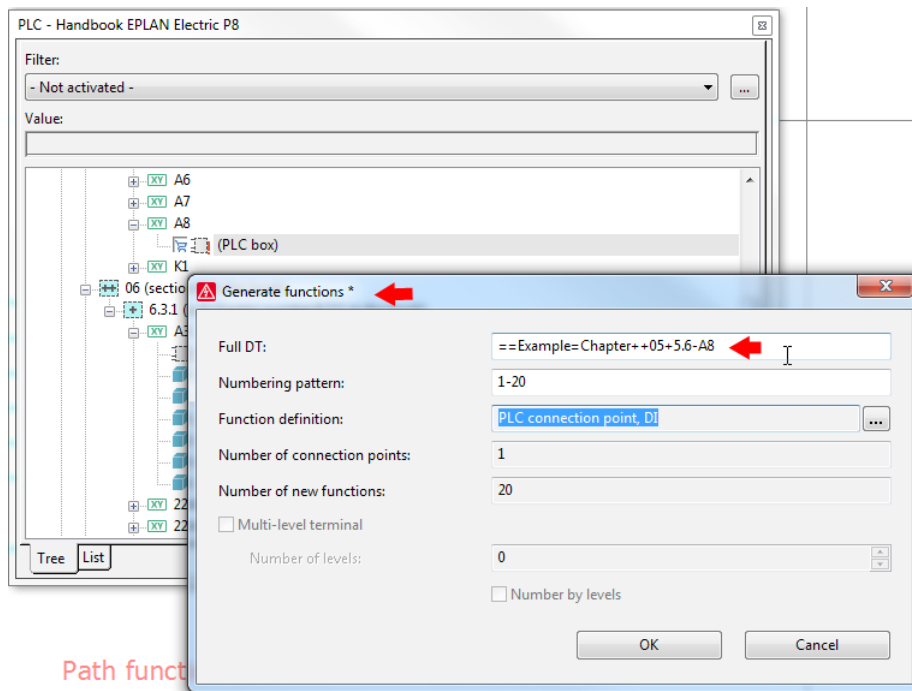


Fig. 5.157 Generating new PLC functions

These functions are generated when OK is clicked. This is also the approach for generating supply connection points for a card. This method creates a complete PLC card in the PLC navigator with correct *PLC addresses* (via the PROJECT DATA/PLC/ADDRESS menu) and their *Connection point designations*, ready to be placed.

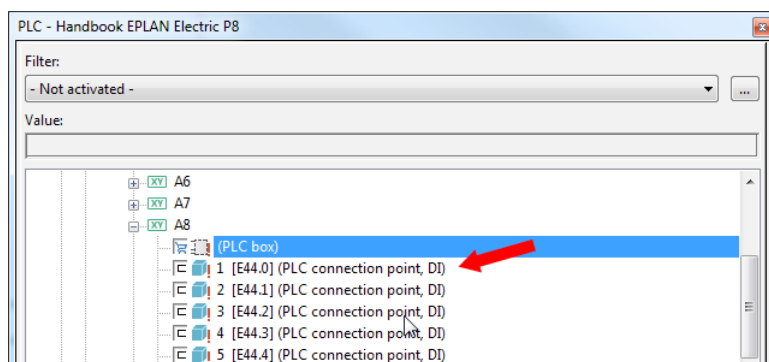


Fig. 5.158 New PLC functions

5.6.8 New device

In contrast to creating individual new functions, the **NEW DEVICE...** item in the popup menu creates a completely new device that already has default function definitions; in the PLC navigator this is, of course, a PLC function, such as an input card.



After you select **NEW DEVICE...** in the popup menu of the **PLC navigator**, EPLAN opens the **PARTS MANAGEMENT** dialog. Only PLC parts can be selected. The selected device is now applied in the PLC navigator by clicking **OK**.

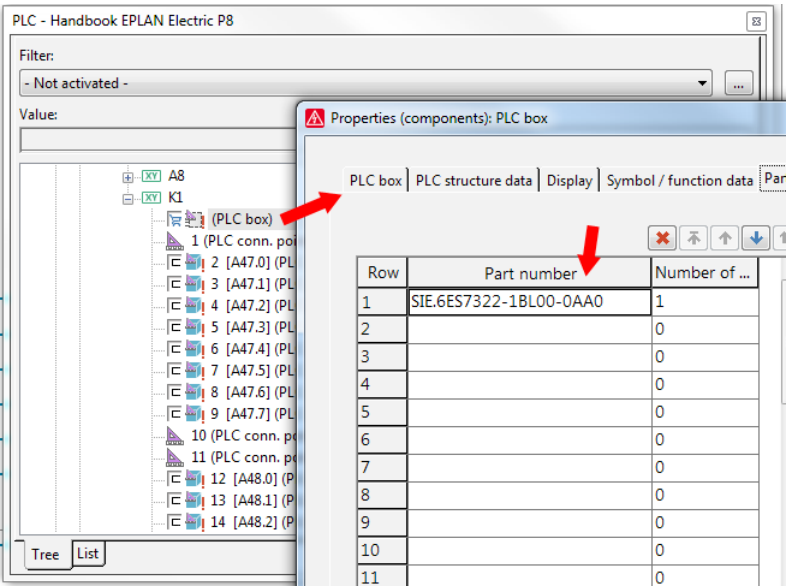


Fig. 5.159
New, unplaced device
in the navigator

EPLAN automatically assigns the next available device tag to the new device in the PLC navigator. The device is then ready to be used in the schematic or ready for its functions to be placed.

5.6.9 View

The **PLC navigator** in EPLAN provides different *views* of the PLC data: **DT-oriented**, **Address-oriented**, **Channel-oriented**, and **Rack-oriented**.

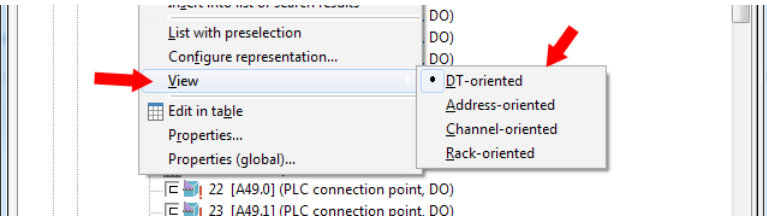


Fig. 5.160
Popup menu selection
of views

5.6.9.1 DT-oriented

The DT-oriented view shows the DT and all functions below it. In the DT-oriented view, the PLC data is listed according to its device tag.

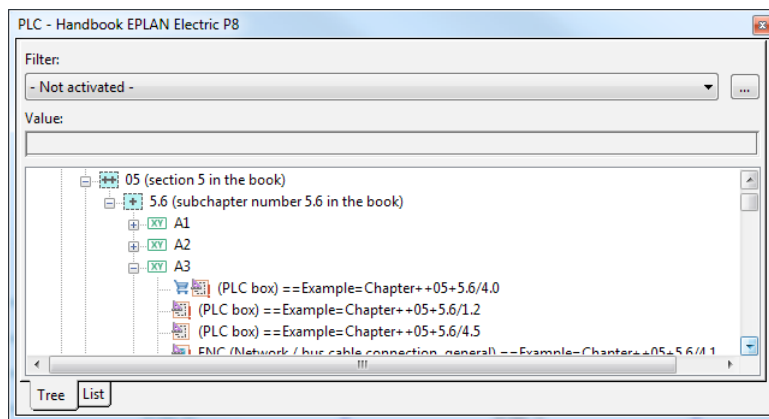


Fig. 5.161 DT-oriented view

5.6.9.2 Address-oriented

In the address-oriented view, the PLC navigator organizes the PLC data by address.

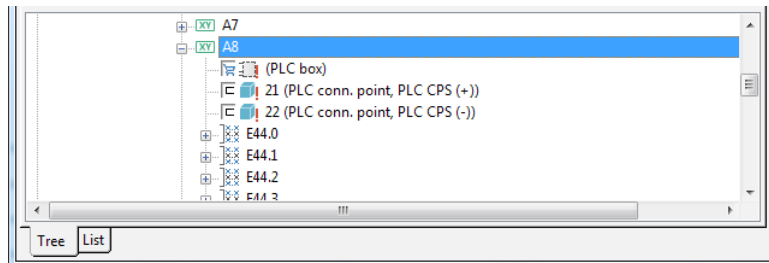


Fig. 5.162 Address-oriented view

5.6.9.3 Channel-oriented

The channel-oriented view shows the channel groups and all of their PLC functions. The channel-oriented view has the advantage of allowing you to select and place all of the functions belonging to a channel “at once”. This ensures that you do not forget any of the functions in the channel group.

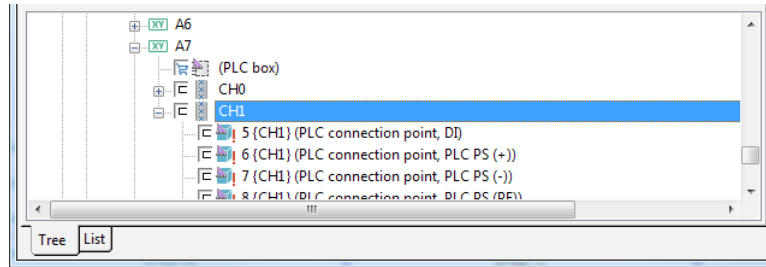


Fig. 5.163 Channel-oriented view

5.6.9.4 Rack-oriented

In the **Rack-oriented** view, the PLC data is displayed based on its assignment of devices to the racks.

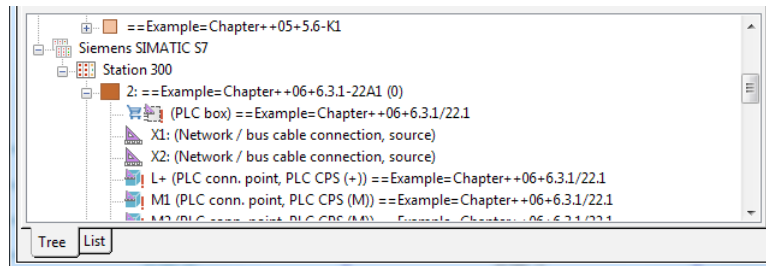


Fig. 5.164 Rack-oriented view

■ 5.7 Part navigator (Devices/Parts)

The **bill of materials navigator** has a view of all devices with or also (configurable in **OPTIONS/SETTINGS/PROJECTS/[PROJECT NAME]/REPORTS/PARTS**) without part numbers that exist in the project. The bill of materials navigator is started via the **PROJECT DATA/PARTS/DEVICES** menu.

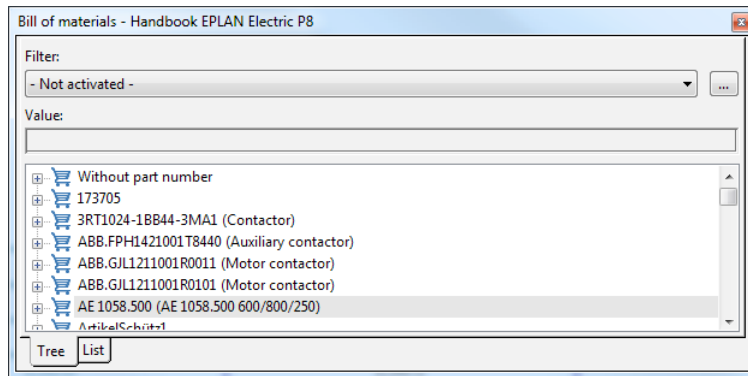


Fig. 5.165 Bill of materials navigator in the tree view

The **bill of materials navigator** offers different functions in its popup menu to make editing the project parts easier.

5.7.1 Add project part

If you select the **ADD PROJECT PART** menu item, EPLAN opens *parts management*, where you can add a part to the project as a *project part*.

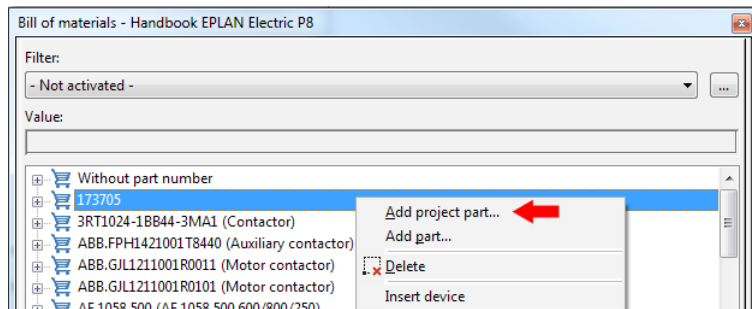


Fig. 5.166 Add project part (popup menu)

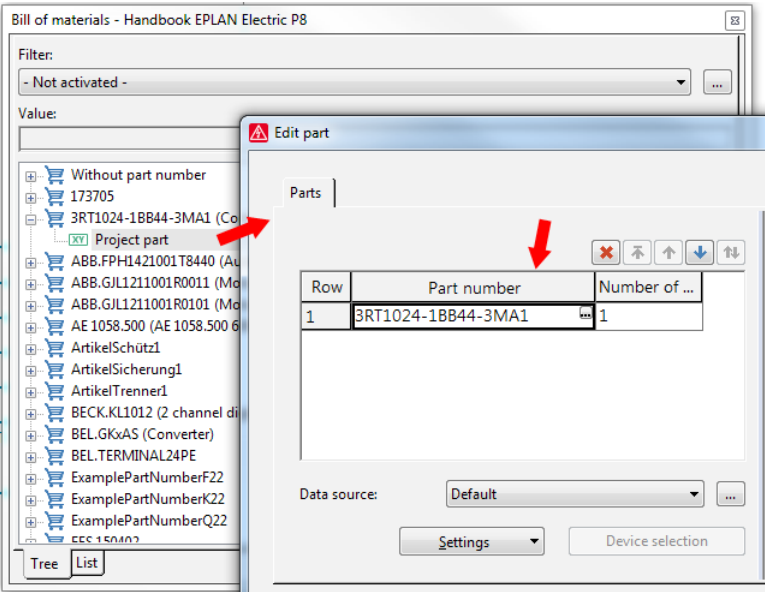


Fig. 5.167 Added project part

Project parts initially do not have device tags. These are normally parts that are shipped as spare or loose parts. Project parts can also be filtered or evaluated.

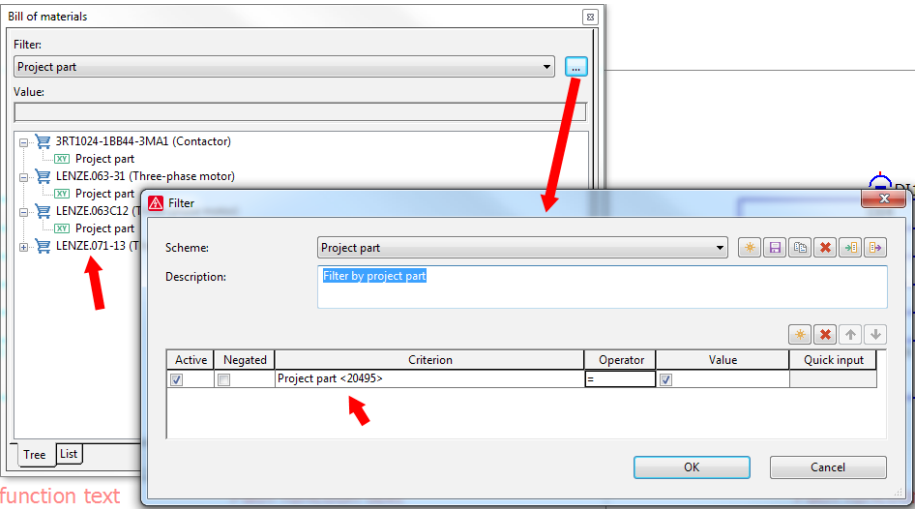


Fig. 5.168 Filter for project part

5.7.2 Add part

You can easily add additional parts to an existing device with the **ADD PART** menu item. Select the device and call the **ADD PART** menu item.

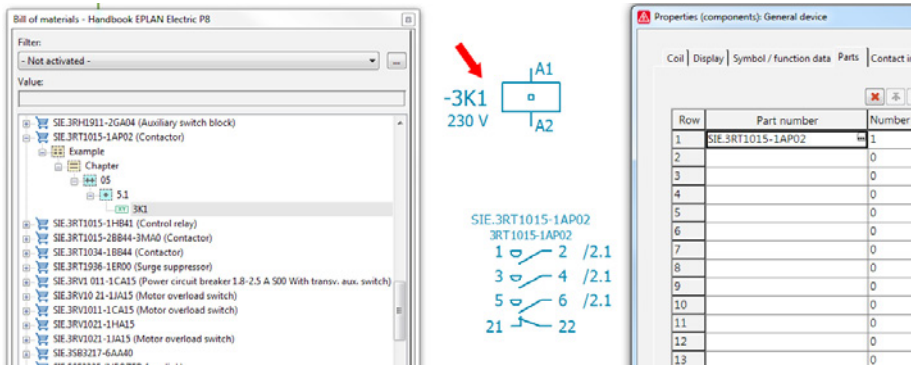


Fig. 5.169 Before running the Add part function

EPLAN starts *parts management* after running the **ADD PART** function. The desired part is chosen, selected and applied to the selected device by clicking **OK**.

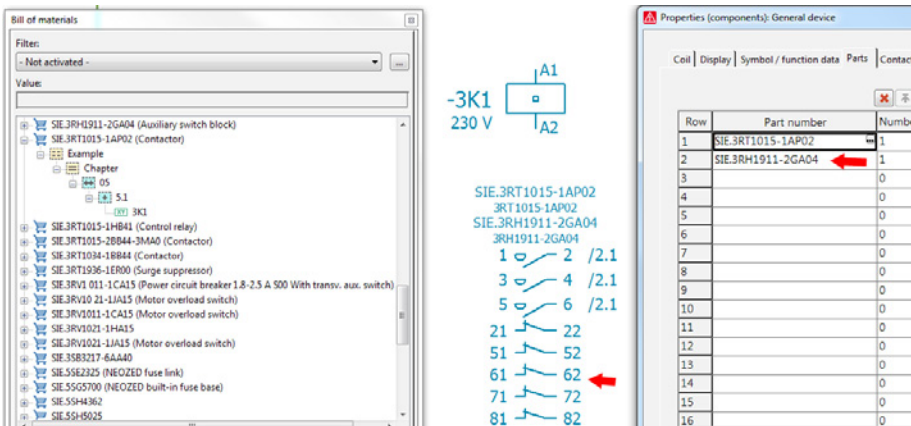


Fig. 5.170 A part was added.

This function of course also works with multiple selections. For example, you could add surge arresters for several contactors “at once” if you “forgot” them.

5.7.3 Insert device

The **INSERT DEVICE** function does just that: it adds a selected device to the graphical editor. Select the desired part, click **INSERT DEVICE** from the popup menu, and then the device hangs at the cursor with an automatic DT numbering, ready to be placed.

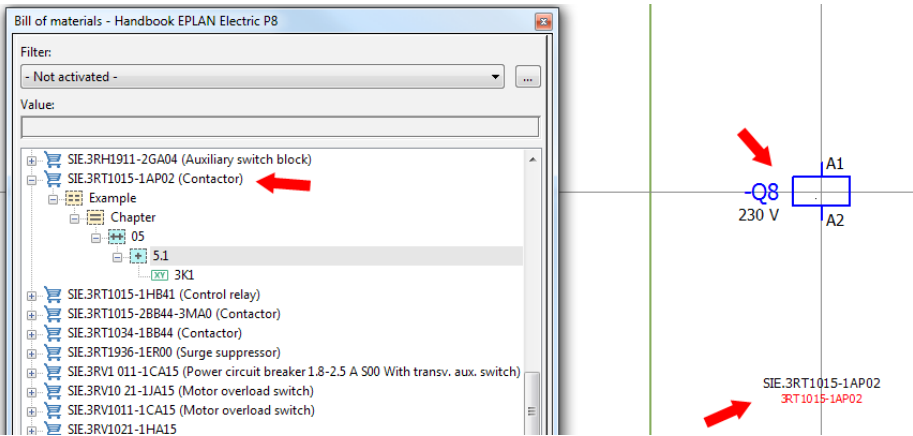


Fig. 5.171 Insert device

5.7.4 Exchange part

This menu item offers the option to exchange a part quickly and easily. Select the part in the **bill of materials navigator** and open the **EXCHANGE PART** menu item. EPLAN then opens **parts management**, where you select the *part to be exchanged* is and then click **OK** to apply it. You may see the **CONFLICT** dialog. Here, you can specify the data to be overwritten or retained.

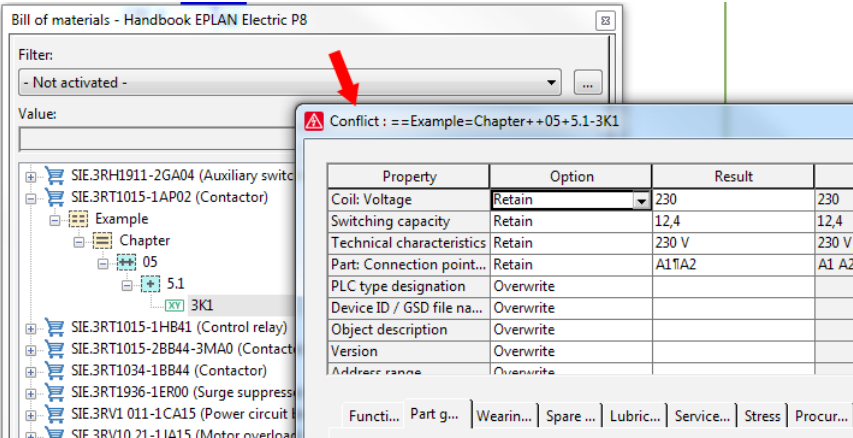


Fig. 5.172 Conflict dialog

EPLAN instantly exchanges this part with the selected one after you click the **OK** button.

5.7.5 Edit part

The EDIT PART function makes it possible to edit the part reference data of a part, to select a different part, or to change the quantity.

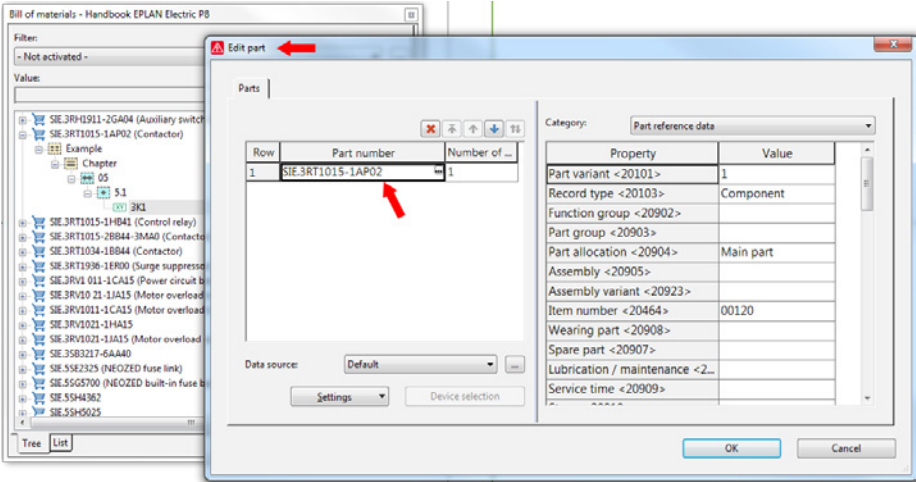


Fig. 5.173 Edit part reference data of a part

5.7.6 Device selection

This menu item belongs to the topic of bulk editing. For example, you can use this to perform a new *contactor selection* retroactively for multiple selected parts data (contactor coils). Select the part and run the **DEVICE SELECTION** menu item.

EPLAN now allows you to perform device selection for every device, one by one. If device selection has not been performed for a device, then it suffices to press the **ESC** key. EPLAN then asks whether the entire action should be cancelled, or whether the selected device should be skipped.

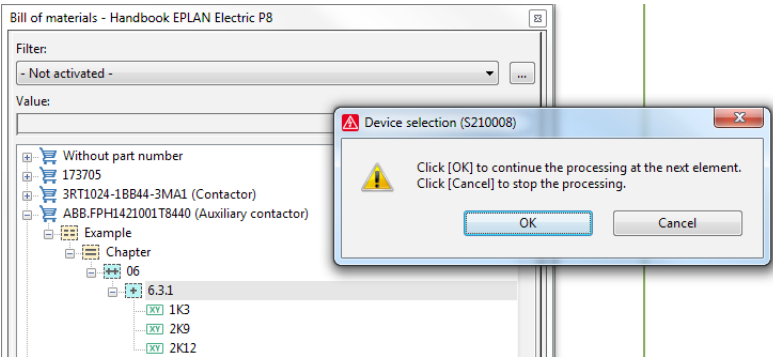


Fig. 5.174 Cancel option in device selection

5.7.7 Assign item number

This menu item opens the **NUMBER ITEMS** dialog. Here, you can implement a new item numbering of the selected item or of all project items.

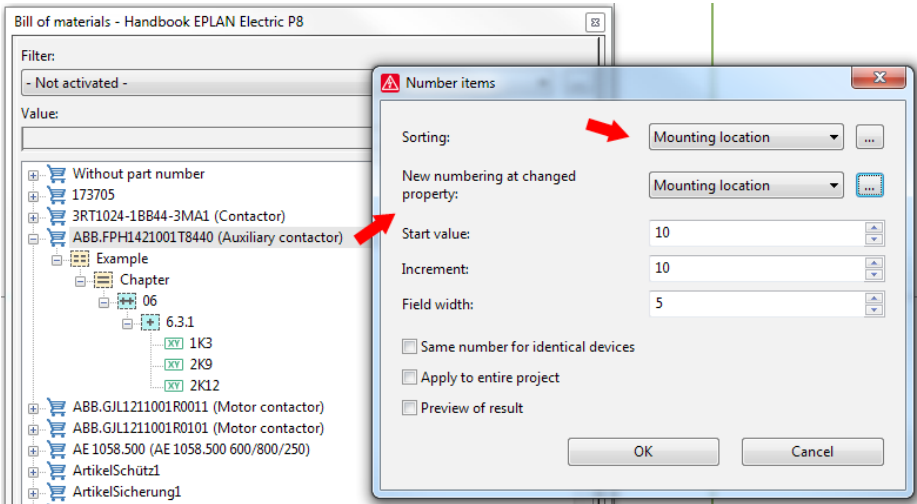


Fig. 5.175 Renumbering items

Filter:	Item num...	Part number <2048...	Number o...	Function t...	Technical ...
- Not activated -					
Value:					
DT <20480>					
==Example=Chapter++02+2.5.1-Q22	00140	SIE.HS-0002	1		400V AC 1...
==Example=Chapter++02+2.5.1-Q22	00150	SIE.AS-0001	1		400V AC 1...
==Example=Chapter++02+2.5.1-Q22	00160	SIE.US-0001	1		400V AC 1...
==Example=Chapter++02+2.5.1-1X1	00170		0		
==Example=Chapter++02+2.5.1-1X2	00180		0		
==Example=Chapter++02+2.5.3-Q1	00010		0		
==Example=Chapter++02+2.6.5-Q1	00010		0	Function t...	63A
==Example=Chapter++02+2.6.5-Q2	00020		0	Function t...	63A
==Example=Chapter++02+2.6.5-Q3	00030		0	Function t...	63A
==Example=Chapter++04+4.1.1-M1	00010		0		
==Example=Chapter++04+4.1.1-M2	00020		0		
==Example=Chapter++04+4.1.1-M3	00030		0		

Fig. 5.176 Result of new item numbering

5.7.8 Synchronize parts data

The SYNCHRONIZE PARTS DATA menu item offers the option of synchronizing the master data of specific, selected parts with parts management. This operation compares the parts data with parts management master data and overwrites the saved parts data in the project if necessary.

5.7.9 Export/Import

This menu item allows for bill of materials data of the entire project to be exported and/or re-imported (file format *.xml). Both functions, however, only work within a project.

5.7.10 View

Like the other navigators, the bill of materials navigator offers fundamentally different viewpoints of the project and its data. Generally, there are four different views: *Part number*, *Part number/Structure identifier*, *Manufacturer*, and *Supplier*.

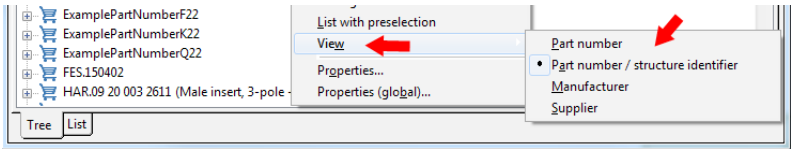


Fig. 5.177 View options

5.7.10.1 Part number

Example of **part number** view. It starts with the part number, followed by the identifier of the device, and finally the actual full DT.

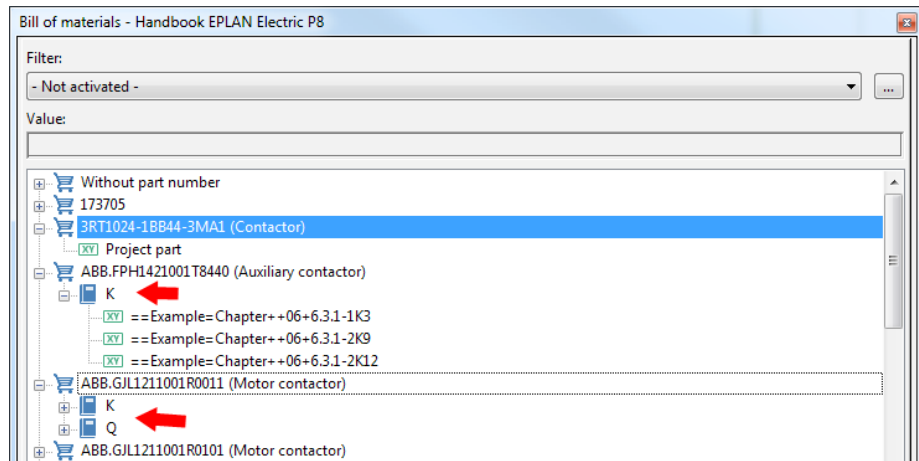


Fig. 5.178 Part number view

5.7.10.2 Part number/Structure identifier

EPLAN realizes the combined view of **part number** and **structure identifier** as follows. It starts again with the part number, followed by the structure identifiers (split by structure), and finally the DT.

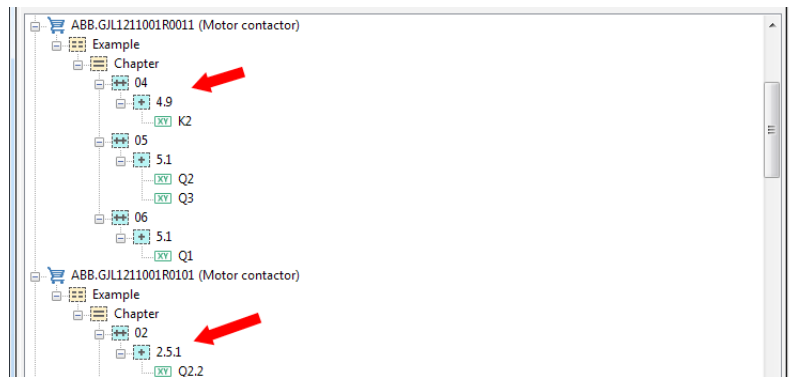


Fig. 5.179 Part number and structure identifier view

5.7.10.3 Manufacturer or Supplier

The **Manufacturer** and **Supplier** views are explained quickly. Depending on the view setting, the manufacturer or the supplier is at the top (both sets of data originate from parts management). This is followed by the part number, identifier and, finally, the full device tag (full DT).

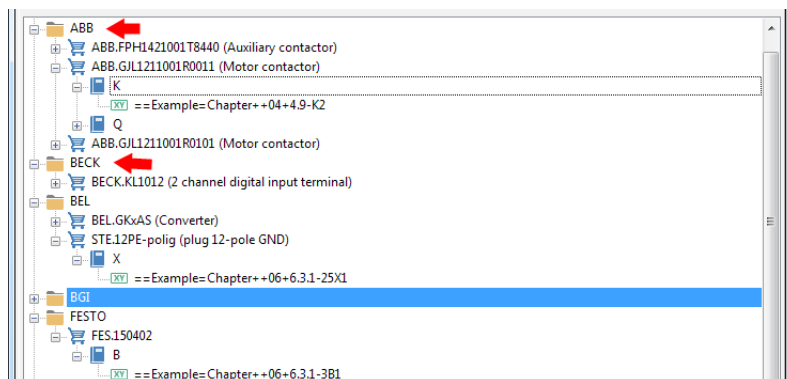


Fig. 5.180 Manufacturer/supplier view

5.8 Macro navigator

The macro navigator provides a complete overview of all macros, their variants and further information in a macro project.



NOTE: So far, the macro navigator serves no other functions in a schematic project.

5.8.1 Macros – a general description

Essentially, macros are a collection of various data that, once processed, is to render project editing more efficient and more convenient. Depending on the requirements, it is greatly advantageous to have maintained and documented macros and/or collections of macros.

Macros can also be created and used without a macro project. This option is explained further in sections 5.8.5 and 5.8.6. But having a macro project affords the user many advantages, which I will attempt to show in the following sections.

5.8.2 Macro project

A macro project is created like any regular schematic project. The differences of a macro project, when compared to a schematic project, are the following: The **Type of project**

<10902> project property is set to macro project; there are no cross-references, and no connections are generated.

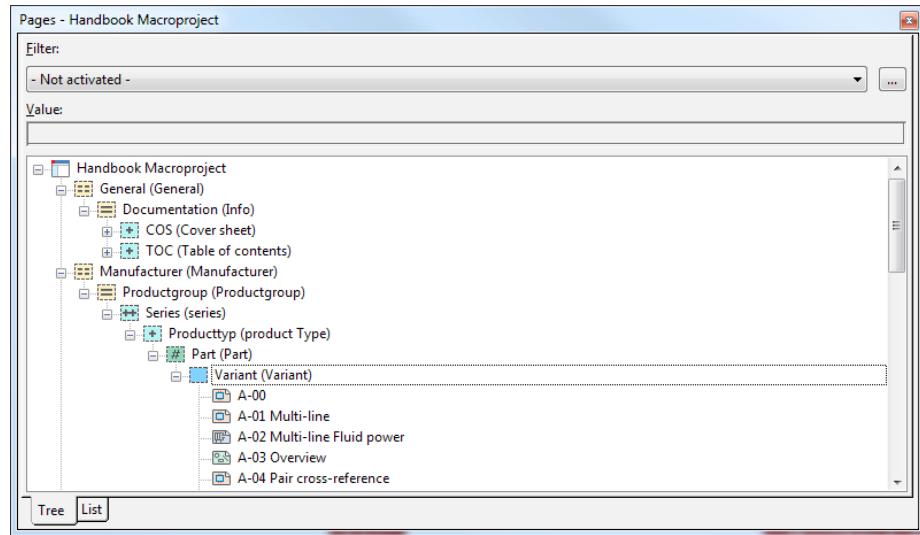


Fig. 5.181 Macro project sample structure

You should consider the structure of the macro project thoroughly before creating and generating macros and their variants. These should be a fit for the structure of the workflows – what is planned with whom and when, etc. If the macro project is built accordingly, maintenance and subsequent extensions do not pose problems, because the structure of the macro project has been well thought out.

“Wild” structures are advised against – having stored “some” macros “somehow” “some-where” in “crazy” project structures. This causes nothing but confusion, and never produces the desired results, such as time and cost savings in project editing.

The next sections refer, always in a general manner, to the approach in a macro project, and illustrate some, but not all, options to be considered in the use of macros in connection with a macro project.

5.8.3 Macro box

The assignment and data of window and symbol macros are defined by means of a macro box. On the basis of special entries, it will be possible to generate from this the macros in desired structures at a later time. Generally, the macro box is inserted into the macro generated.

Depending on the setting, this macro box can be inserted later on in a schematic project or when using the macro, for example. This setting is found in the **OPTIONS/SETTINGS/PROJECTS/[PROJECT NAME]/GRAPHICAL EDITING/GENERAL** menu.

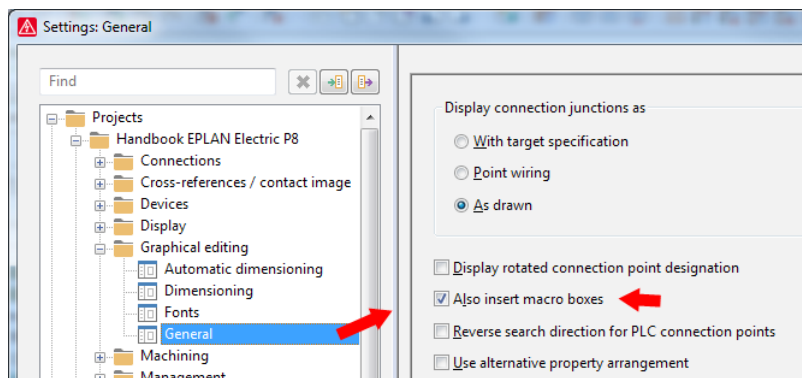


Fig. 5.182 Setting for inserting macro box

The macro box should always be inserted as well (recommended). For any subsequent editing or adjustment of data regarding the macros already used and inserted, this will be nothing but beneficial. For example, an edited macro variant can be synchronized (replaced) quite easily later on, because EPLAN remembers the macro's assignment.



NOTE: There are no macro boxes with page macros. Here, the data is set in the page properties.

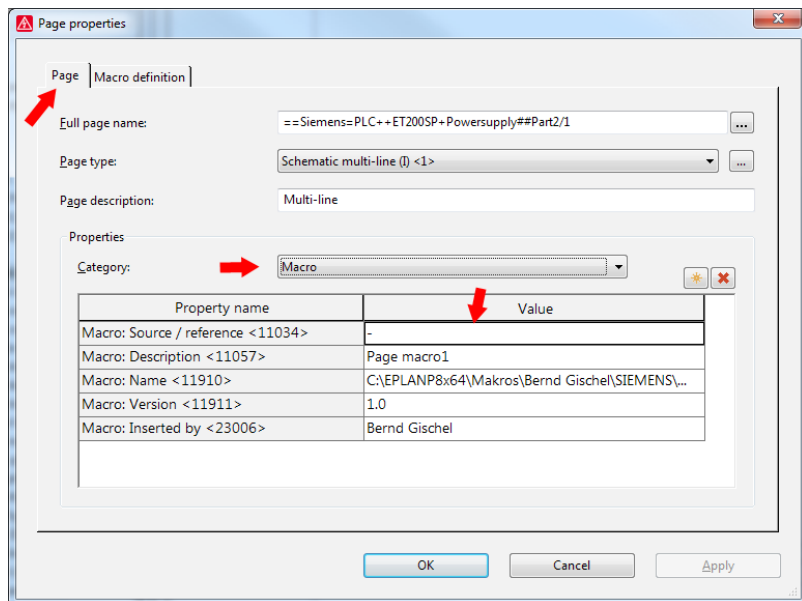


Fig. 5.183 Page properties data for page macros

5.8.3.1 General information on macro boxes

The macro box itself is located on a specific layer: EPLAN308. As the user knows, this can be hidden using the layer management tools, such as Print.

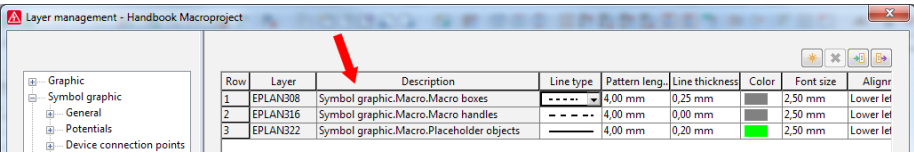


Fig. 5.184 Layers of a macro

It is also possible to let EPLAN remove from a project the inserted macro boxes in one go, using the compression function and a suitable scheme (PROJECT/ORGANIZE/COMPRESS menu).

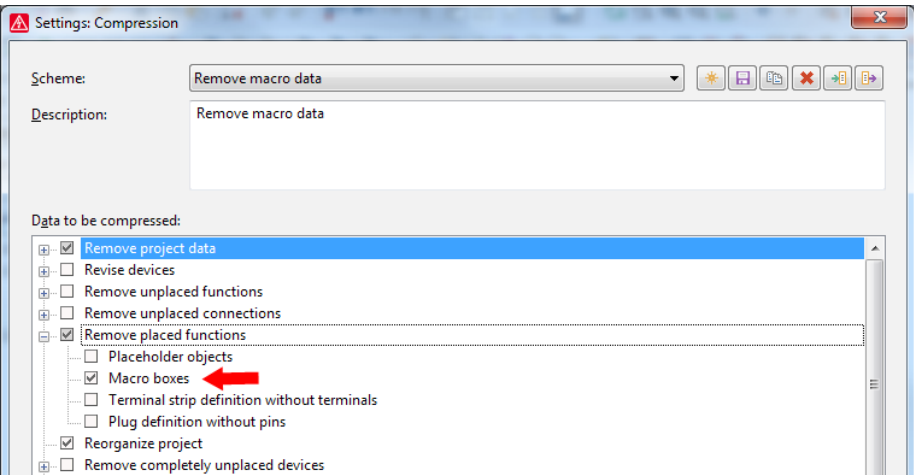


Fig. 5.185 Compression function

5.8.3.2 Macro box tab

Once the circuit, partial circuit or part (for example, an overview or panel layout representation) has been drawn, the macro box can be inserted.

The examples shown constitute the easiest representations without any further technical background. The primary purpose here is to illustrate the way in which the macro project/macro box and macro navigator are used.

Macro boxes are inserted via the INSERT/BOX/CONNECTION POINT/MOUNTING PANEL/MACRO BOX menu.

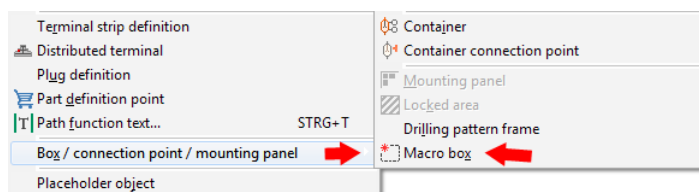


Fig. 5.186 Calling Insert macro box

The macro box is drawn across the objects like a black box.

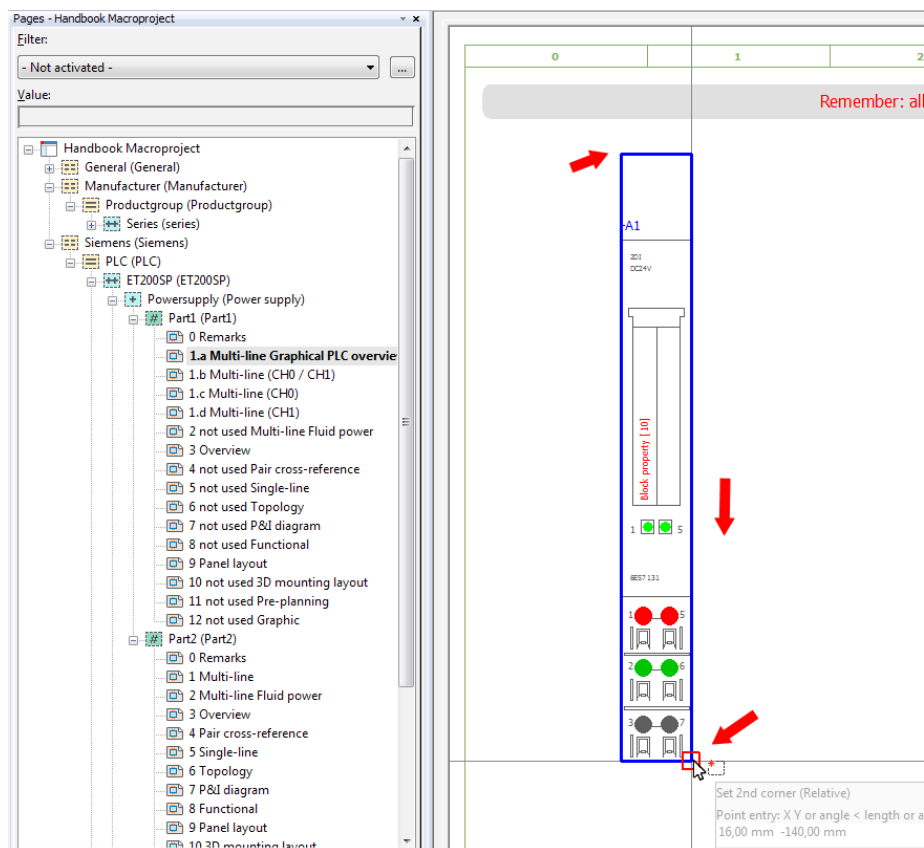


Fig. 5.187 Assigning objects to the macro box

After the assignment (bordering) of all objects, the second corner can be clicked. Now, another macro box can be inserted, or the action can be canceled.

Then, you can left-double-click the macro box. EPLAN opens the symbol properties of the macro box.

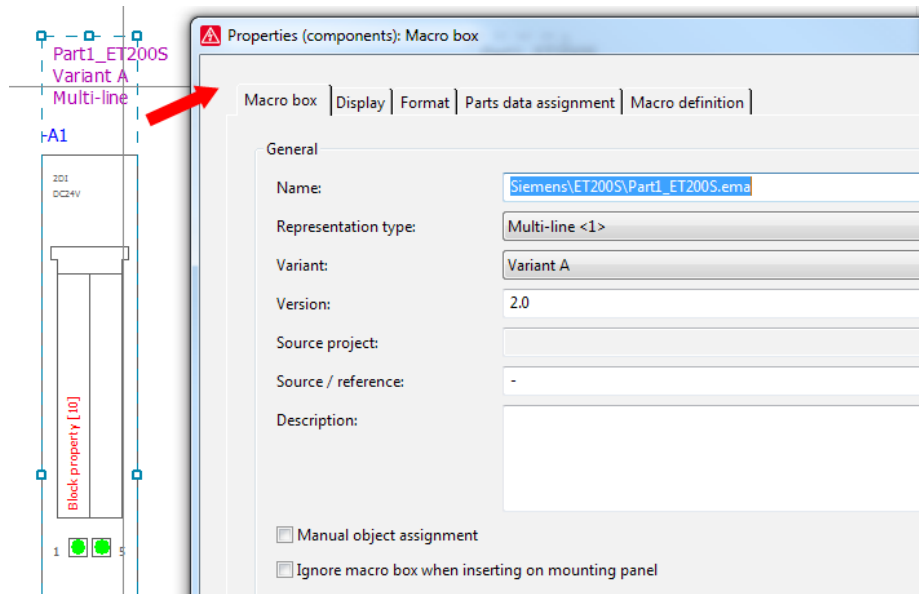


Fig. 5.188 Macro box dialog Macro box tab

5.8.3.2.1 Name, variant and more

Now, you can configure the settings for the macro on this tab. A name (macro name) is required. The other entries can, but need not, be made.

All fields should be filled as carefully as possible (recommended). This way, it will be easier to trace changes later on.

Path information can be included in the name (macro name). By default, EPLAN uses the default macro system directory. A specific structure (directory structure) can be entered here as well. This does not have to be an existing structure. EPLAN will create it automatically when generating the macro.



If no file extension is defined, EPLAN will use the file extension *.ema (window macro) automatically. If you specifically want a symbol macro output, then you must use the file extension *.ems with the macro name.

In the case of page macros, information, such as the macro name or any deviating directory for saving the page macro, is taken from the page properties, because there are no macro boxes here (i.e. page macros).

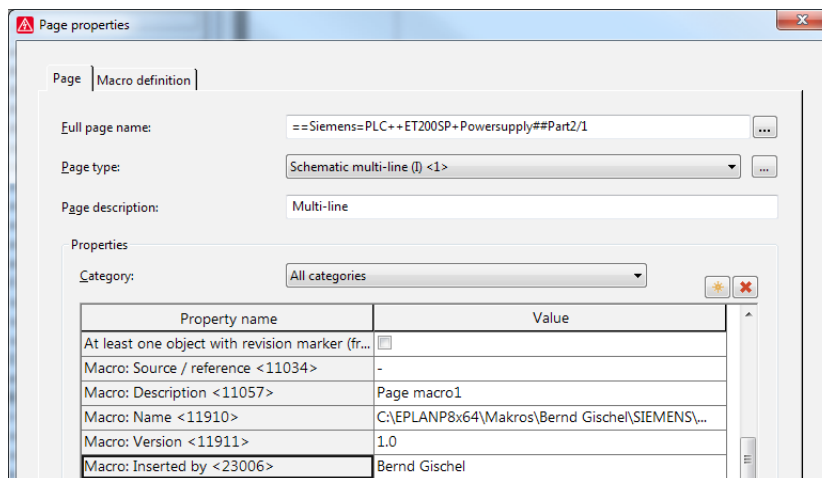


Fig. 5.189 Page properties dialog

5.8.3.2.2 Handle active

The handle is usually in the upper left. Using this option, the handle can be modified in any manner whatsoever. For this purpose, the option is enabled, which means you can make manual changes to the X and Y positions. EPLAN, then, indicates this “move” by means of a special symbol.

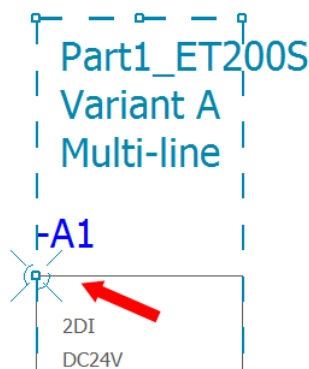


Fig. 5.190 Handle moved symbol

5.8.3.2.3 Manual object assignment

The **MANUAL OBJECT ASSIGNMENT** option allows for objects outside the actual macro box to be incorporated into the extent of the macro box.

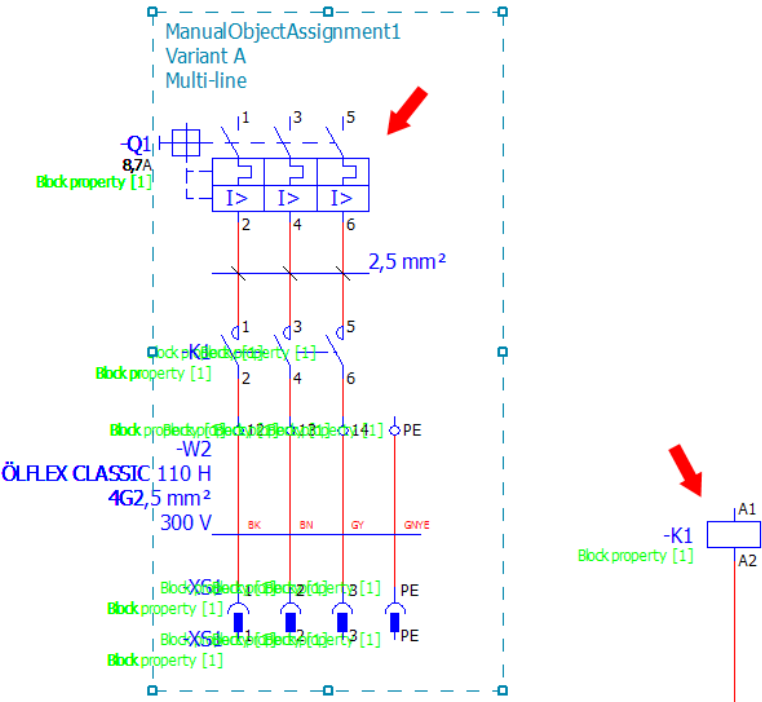


Fig. 5.191
Macro box without
manual object
assignment

To integrate the exterior objects with the macro box, you must click the macro box, then right-click to open the popup menu of the macro box, followed by the **ASSIGN OBJECTS TO MACRO BOX** menu item.

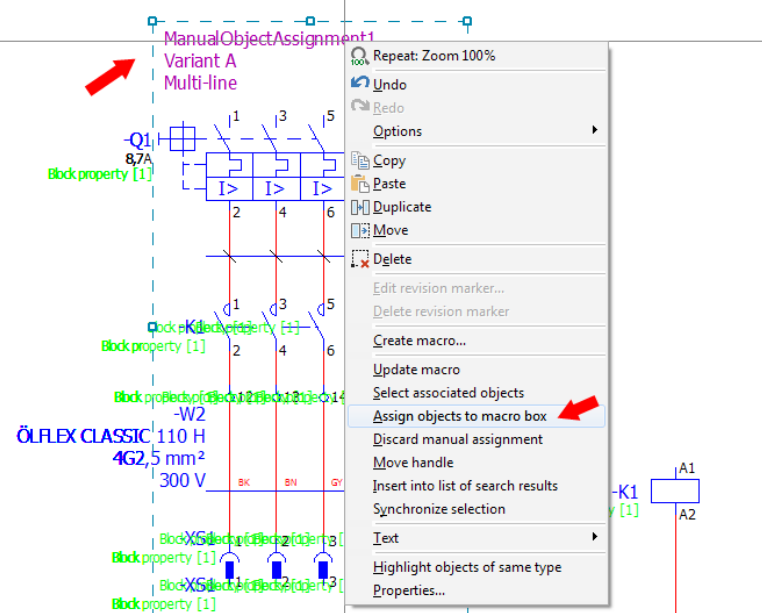


Fig. 5.192
Macro box popup
menu

Now, select all elements using the mouse, then close this action with the second clicked point. All objects are assigned to this macro box now.

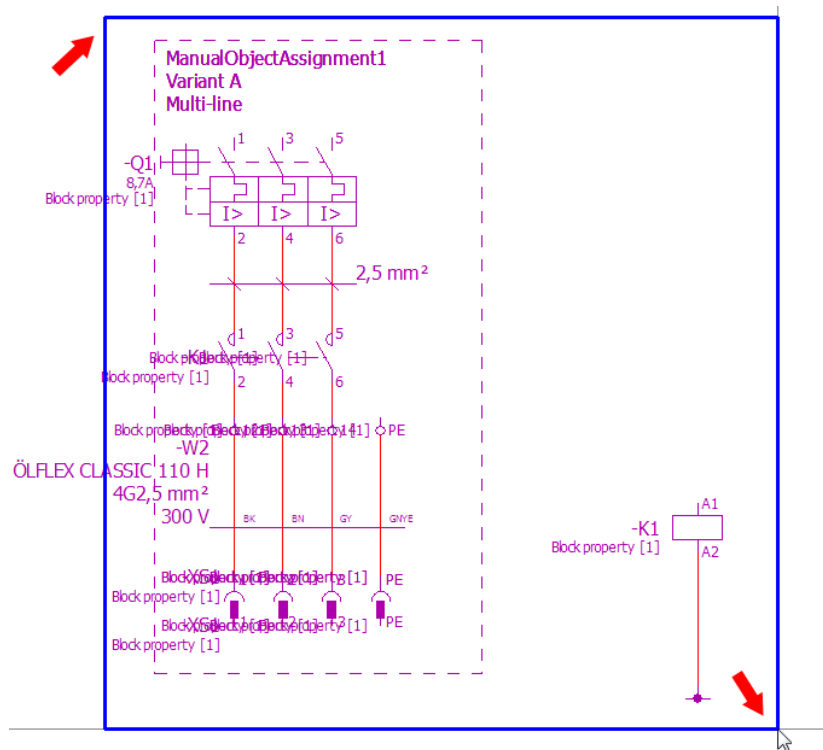


Fig. 5.193 Assigning exterior objects to the macro box

To verify, you can call the popup menu of the macro box and click the **SELECT ASSOCIATED OBJECTS** menu item. EPLAN selects the corresponding objects.

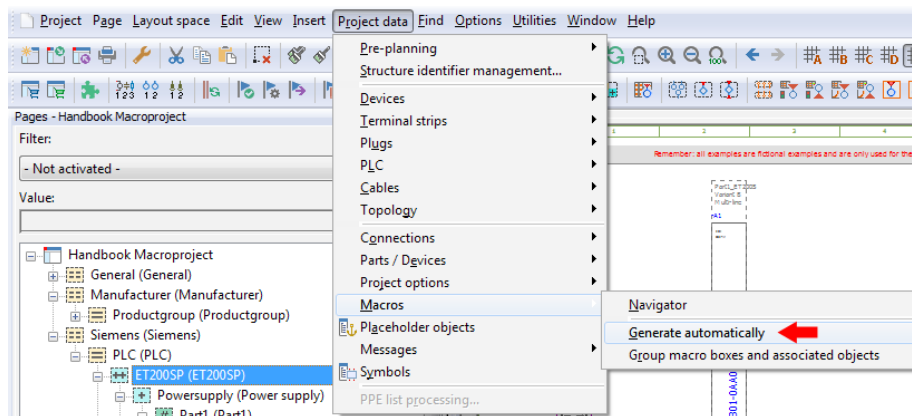


Fig. 5.195 Generating macros automatically

Before the actual generation, EPLAN displays a prompt as to whether only the selected macros or all macros in the macro project are to be generated.

When you click the YES button, EPLAN generates the macros automatically in the directories specified. This concludes the process of creating and generating macros. Now, these macros are ready for use.

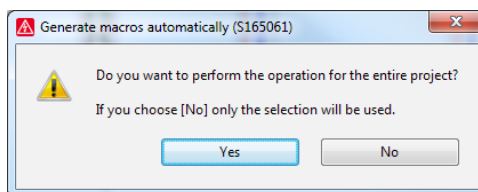


Fig. 5.196 Prompt

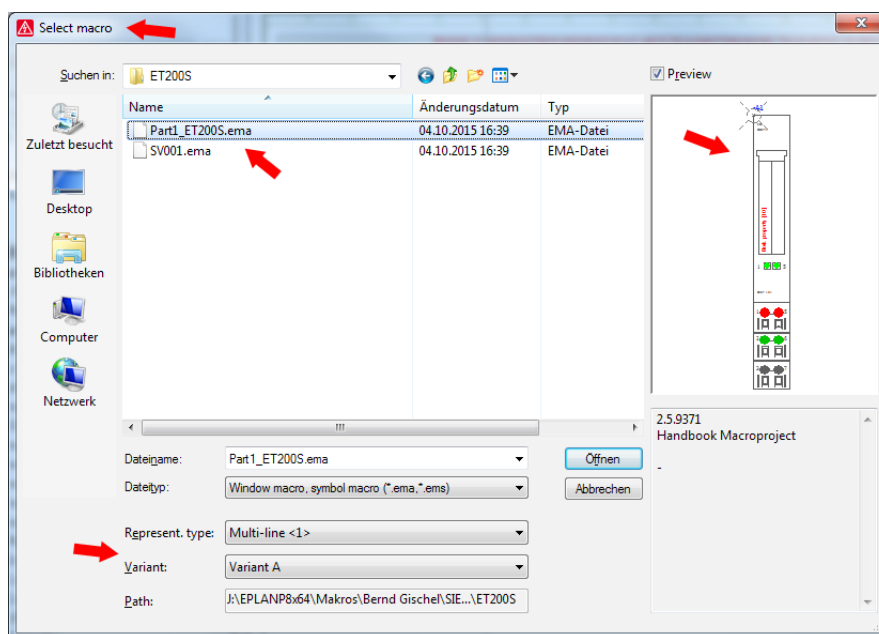


Fig. 5.197 Using a macro

If the macro is changed (in the macro project, of course!), you can have the macros re-generated and, subsequently, updated easily in the schematic project used. In other words, there is no need to delete anything, re-insert the macro or re-number the DT, etc.

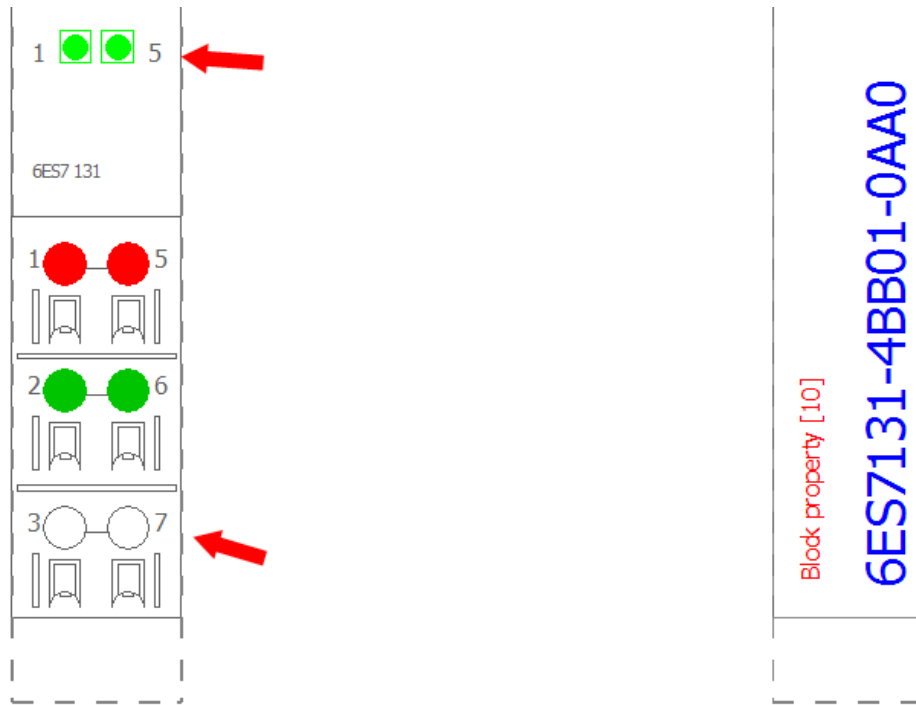


Fig. 5.198 Modified macro in the macro project

To update the macro, click the macro box and call **UPDATE MACRO**. EPLAN instantly changes the macro, showing it in the new version. This requires, of course, that the macro box has always been inserted as well (this option can be set in the **OPTIONS/SETTINGS/PROJECTS/[PROJECT NAME]/GRAPHICAL EDITING/GENERAL** menu).

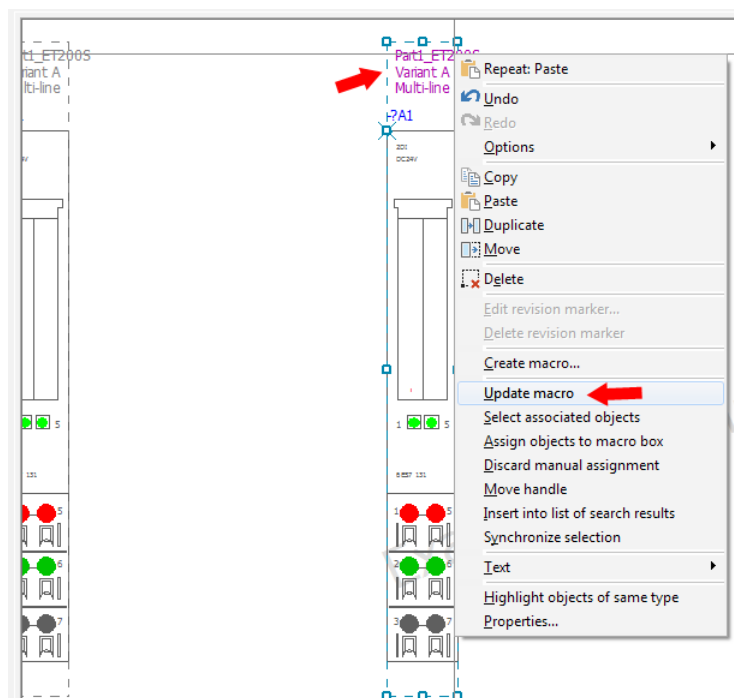


Fig. 5.199 Macro box popup menu – Update macro

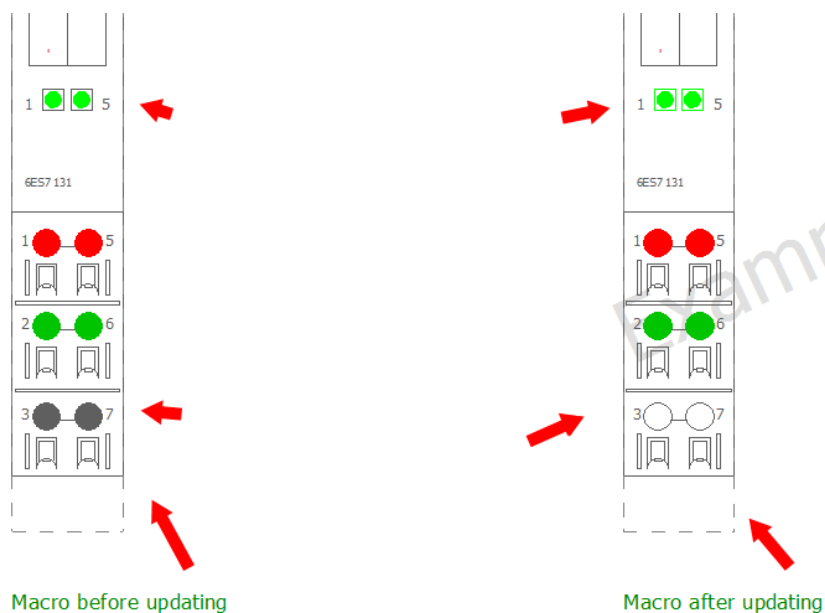


Fig. 5.200 Successfully updated macro

5.8.5 Macros in general (without macro project)

Of course, apart from the macro project and/or generating the macro automatically, you can also create and save macros manually in a purely conventional manner. The following sections deal with this method, but it needs to be said that it is better to create, maintain and generate macros in a macro project, and to use only these macros later on.

Furthermore, there is always the option of generating macros from schematic projects, transferring them to a macro project, and then generating and maintaining them in the manner described.

5.8.5.1 Types of macros

EPLAN has different types of macros. These can be window, page and symbol macros. Special macros are macros with value sets. These can be window and page macros. When using or creating macros (exception: macros with value sets), it generally makes no difference whether it is a window macro on a multi-line page or a macro on a graphical page.

5.8.5.2 Window macros

Window macros are (can be) the smallest partial circuits in EPLAN. Window macros can include single or multiple devices and objects within an area, or several items within a page.

To create a window macro, you first use the mouse to select the associated devices on the respective page. These may be individual devices or unrelated parts circuits on a project page. These are “gathered” by selecting the first device, and then selecting the other devices that are to belong to the window macro while holding down the **CTRL** key.

You then press the shortcut key **CTRL + F5** or use the **EDIT/CREATE WINDOW MACRO/SYMBOL MACRO** menu to start the function. EPLAN opens the **SAVE AS** dialog. First, you must define a meaningful file name for the macro. The macro directory is usually the company-specific macro directory, but this can be changed if desired.

Window macros are inserted into a page using the **M** key or the **INSERT/WINDOW MACRO/SYMBOL MACRO** menu. The **SELECT MACRO** dialog opens. Here you can select the desired *variant* or *representation type* from the default directory (or a different macro directory by selecting a different directory in the *Search in:* selection field). When you click **OPEN**, the macro will hang on the cursor and can be placed anywhere on the page.

To summarize, window/symbol macros are only possible on *one* (the same) page but with different variations, such as the gathering of objects that do not all have to be within a single window. A window/symbol macro can also be the contents of the entire page (all objects on the page). Window macros have the file extension **.ema*.

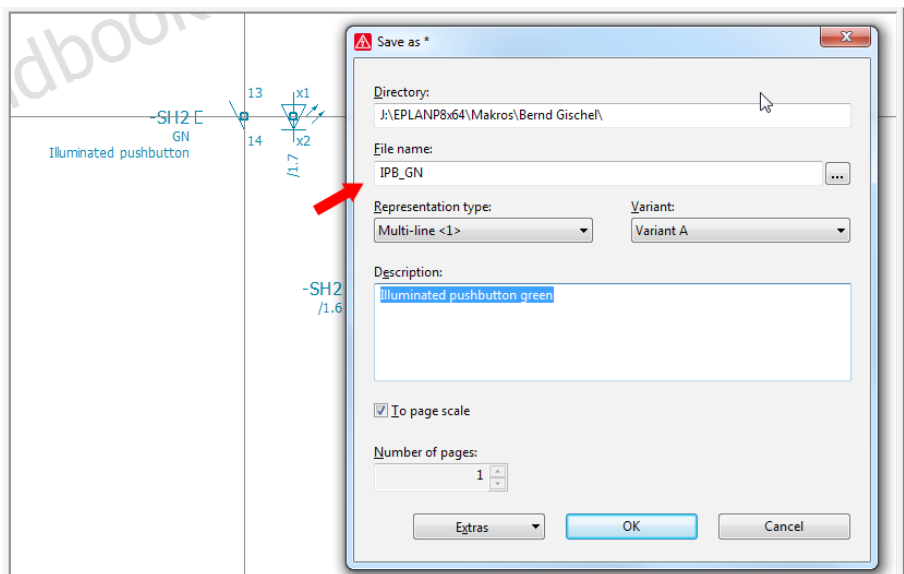
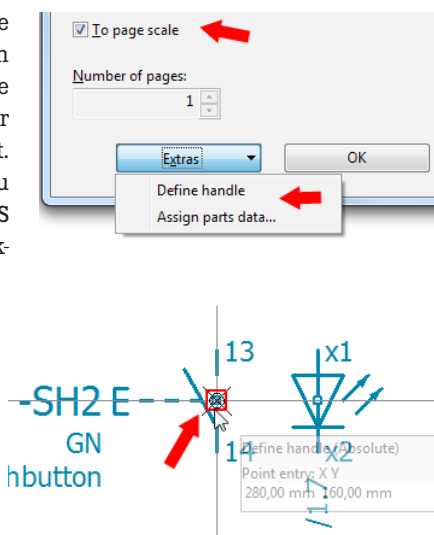


Fig. 5.201 Saving a macro

5.8.5.2.1 Define handle setting

The **MOVE HANDLE** menu item lets you save a (separately created) handle (base point) with the macro. A handle means that when the macro is inserted, it “hangs” on the cursor crosshairs using this handle as the base point. When you click the **MOVE HANDLE** menu item EPLAN temporarily closes the **SAVE AS** dialog, and you can define the handle by clicking with the left mouse button.

The cursor changes when doing this, as can be seen in Fig. 5.203. It is a good idea to enable the **OBJECT SNAP** setting here. Once the handle has been defined, EPLAN returns to the **SAVE AS** dialog, and the macro can be saved.



The **SAVE AS** dialog and the **EXTRAS** button contain other settings related to the macro.

Fig. 5.202
Extras settings

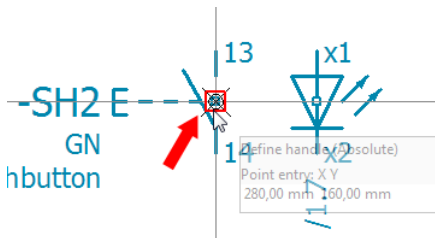


Fig. 5.203
Defining new handle

5.8.5.2.2 Define representation type setting

The **Representation type** selection allows you to define the representation type for the macro.

The individual representation types are usually defined via the devices at the symbol (**SYMBOL /FUNCTION DATA** tab). However, EPLAN allows a different representation type to be defined for each individual macro. This way, you can save many representations under one macro name.

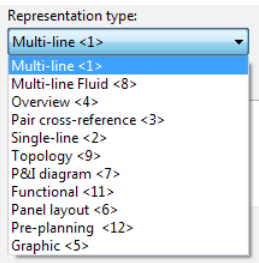


Fig. 5.204
Representation types

It is possible to have a macro for the **multi-line** representation type, one for the **part placement** and one for the **single-line display**. When selecting a macro, EPLAN chooses the right representation type based on the page type defined. Of course, this must be present in the macro.

5.8.5.2.3 Define variant setting

The **Variant** selection box provides another way of working more effectively.

This box allows different variants (so far, sixteen variants A to P) to be stored within a macro. This just means that the same macro name (file name of the macro) can have different content. Depending on the variant used and in connection with the representation types, you can have up to 176 possible variants (macro variants) within one macro. For example, you could save different macro variants of a PLC card in one macro.

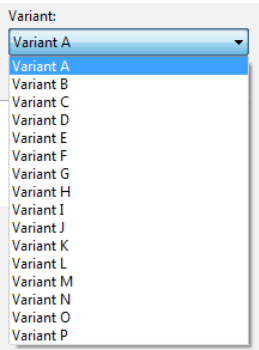


Fig. 5.205
Variants

5.8.5.2.4 To page scale setting

In addition to all the other settings for a window macro, there is also the **To page scale** option. If this option is activated, then EPLAN reduces or enlarges the macro to suit the page scale when it is inserted into a page.

5.8.5.3 Symbol macros

Symbol macros are an additional type of macro. Generally, symbol macros are completely identical to window macros in terms of handling and creation. Further explanation will therefore not be provided. Symbol macros have the file extension **.ems* and are also inserted via the **INSERT /WINDOW MACRO /SYMBOL MACRO** menu.

5.8.5.4 Page macros

Page macros are the counterpart to window macros. Page macros can comprise one or several pages within an EPLAN project. Page macros are created using the **CTRL + F10** keyboard shortcut (at least one page must be open and the user must be on this page), or using the **CREATE PAGE MACRO** popup menu item on selected pages in the **page navigator**.

The familiar **SAVE AS** dialog, as described in the 5.8.5.2 “Window macros” section, is displayed. The *file name* and *description* can be entered here. **No** handles, **no** representation

types, and **no** variants are possible with page macros. However, page macros store all the information relating to a page or pages (page properties). This includes, for example, the structure identifiers or the form pages stored in the page properties.

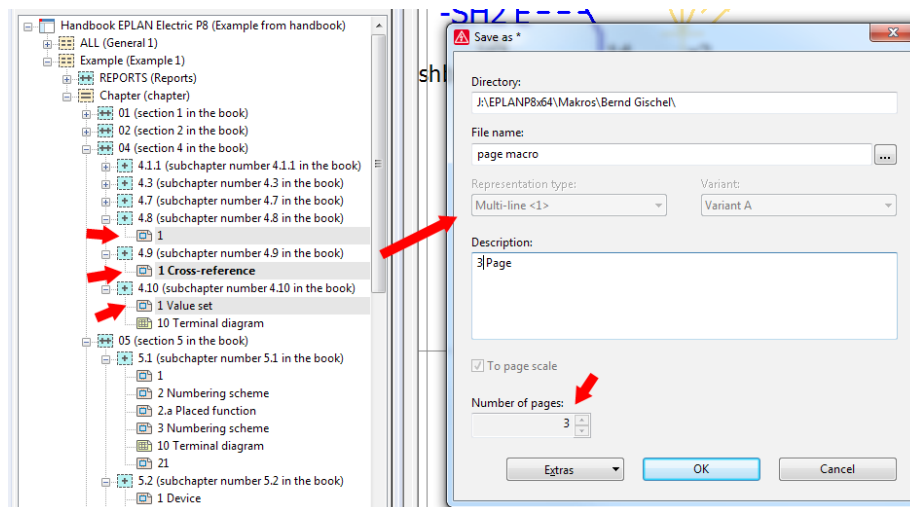


Fig. 5.206
Generating page macro

In addition to the individual page macro (directly on the page), in the **page navigator** it is also possible to write multiple pages (even pages that are not related) into a page macro. You open the **page navigator**, select the desired pages and use the right-click popup menu to call the **CREATE PAGE MACRO** command.

You can use custom keyboard shortcuts for inserting page macros, e.g. **CTRL + ALT + F10**, or call up the popup menu in the **page navigator** and select the **INSERT PAGE MACRO** command. EPLAN then opens the **SELECT MACRO** dialog. Here you select the desired macro and click **OPEN** to apply it.

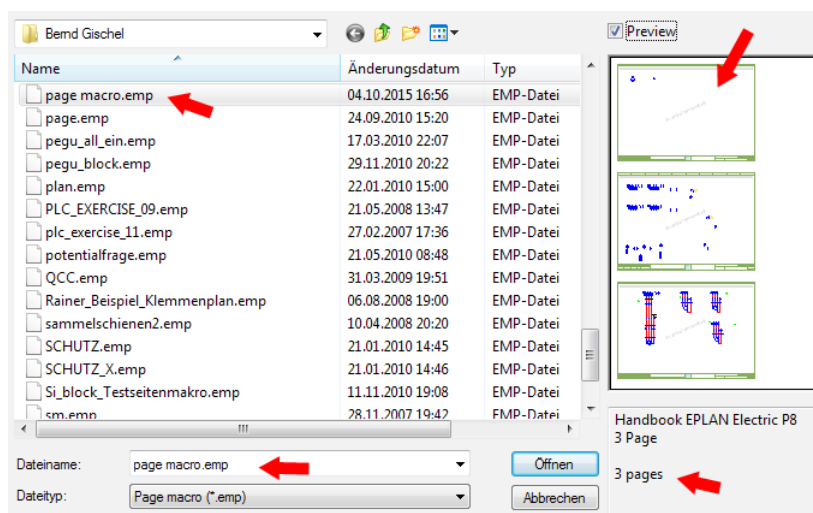


Fig. 5.207
Inserting page macro

If the preview is switched on in this dialog, then all pages are displayed in a small preview. In addition to the preview display (which has a fixed size), the **GRAPHICAL PREVIEW** window can also be permanently displayed (**VIEW** menu). The advantage of this window is that the size can be changed, and you can see the details better.

After applying the macro, EPLAN opens the **ADAPT STRUCTURE** dialog. Here, the page(s) can be stored in the project and sorted into the existing page structure. At this point, it is, of course, possible to adjust the structure of the pages as desired.

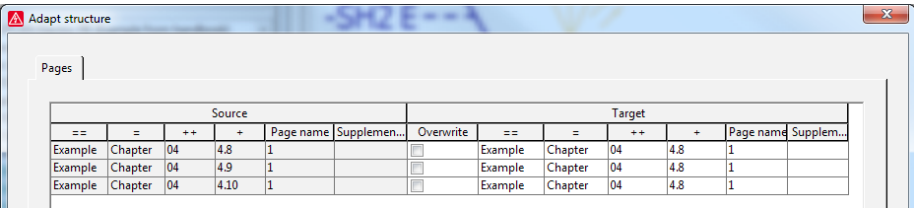


Fig. 5.208 Adapting structure

To summarize, page macros are macros containing one or more pages and their page information (page properties). Page macros have the file extension **.emp*.

5.8.6 Macros with value sets (without macro project)

Macros with value sets are a special function type in EPLAN Electric P8. These are usually window macros equipped with additional functionality that can make project planning much easier. In addition to the basic macro (partial circuit) and its properties, such as technical characteristics, part numbers, etc. of the various objects, this type of macro also contains additional sets of properties. These additional properties are called **Value sets**. These value sets are activated via a specific symbol – the placeholder object. This symbol is used to switch between value sets.

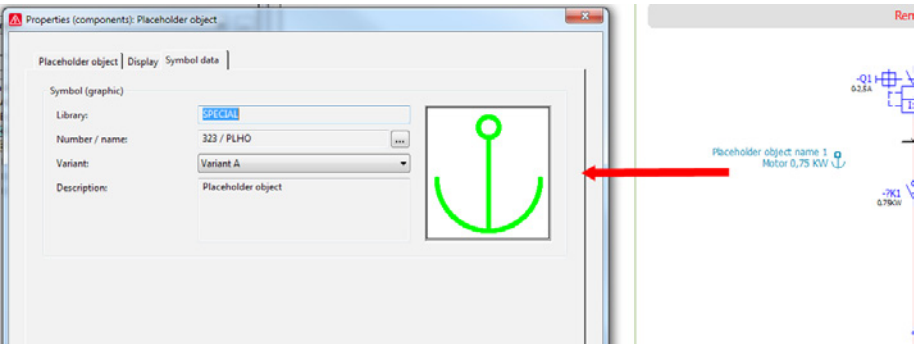


Fig. 5.209 Placeholder object symbol

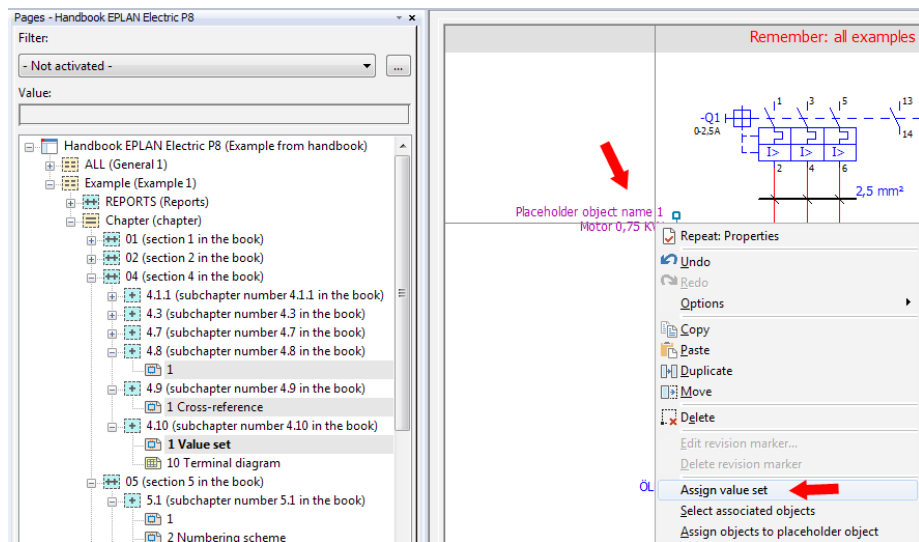


Fig. 5.210
Assigning value set

For example, all devices of the window macro could be equipped with “Siemens parts”. A second value set could then contain “Moeller parts data” for all devices. The value set can then be used to switch between Siemens and Moeller parts data for the same macro with a mouse click.

Once these macros and the properties they contain have been developed and tested, they provide a source of error-free data.

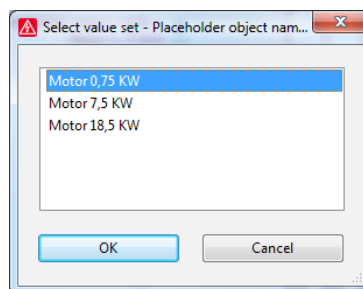


Fig. 5.211
Selecting value set

5.8.6.1 Placeholder object

The most important object in a macro with value sets (subsequently called a value set macro here) is the *placeholder object*. A placeholder object can be inserted via the INSERT/PLACEHOLDER OBJECT menu, making it easy to later switch between value sets. The symbol for a placeholder is an anchor. Since the placeholder object is a symbol, it has settings and options similar to other symbols.

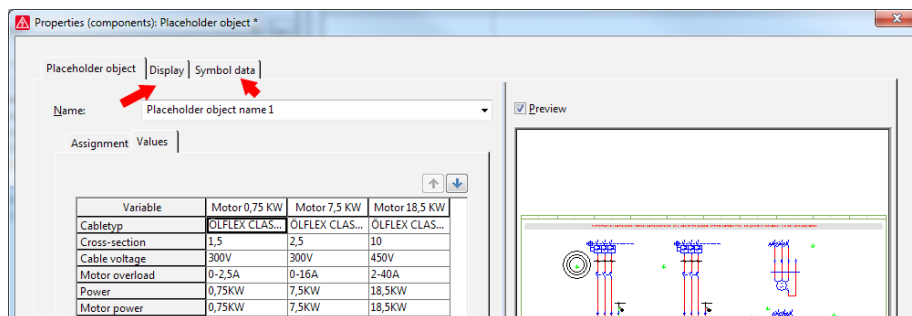


Fig. 5.212
Placeholder object
symbol properties

5.8.6.2 Value set

A *value set* is a collection of variables of selected objects stored in a window macro. Value sets are managed in a type of table “behind the placeholder object” (*Placeholder object* tab) and, in addition to all the device properties (filled or empty), they contain additional information such as the actual values or the variables for the values.

5.8.6.3 Variables of a value set

Variables bring a value set to life. Without these variables, there would be no table for the actual values used for switching the value set. Every property of a device can be provided with any desired variable name. Variable names are always surrounded by the “less than (<)” and “greater than (>)” characters on the keyboard. For example: **<Variable name>**.

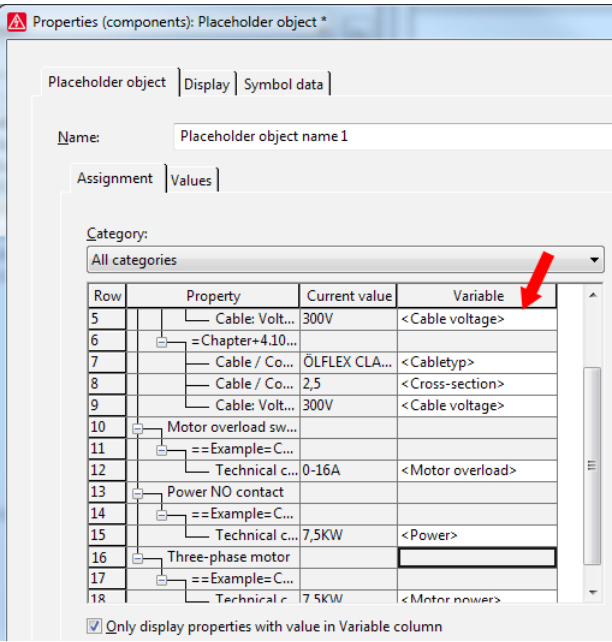


Fig. 5.213 Example of variables

Apart from one exception, all characters can be used. Square brackets are the exception. These have a special purpose within a variable. They are used to define a line break. For example, a variable **<Variable name [12]>** is split at the twelfth position (line break) and the words are simply split at this twelfth position. If the additional option 1 is inserted into the square brackets: **<Variable name [12,1]>**, then a split also occurs at the twelfth position (i.e. a maximum of twelve characters) but an attempt is made to recognize the ends of words. In other words, the words are not “torn apart” but remain intact, and line breaks are only inserted between the words.



NOTE: The square brackets **must** be entered directly after the variable name. Spaces are allowed, but then the defined line break option does not take effect.

5.8.6.4 Create a macro with a value set

Only a few steps are needed to create a value set macro. First, you create the partial circuit with all the required or desired devices and their associated functions such as part numbers, technical characteristics, or function texts. Then you insert the **placeholder object** via the **INSERT/PLACEHOLDER OBJECT** menu. There are several ways of assigning objects to a placeholder object in EPLAN.

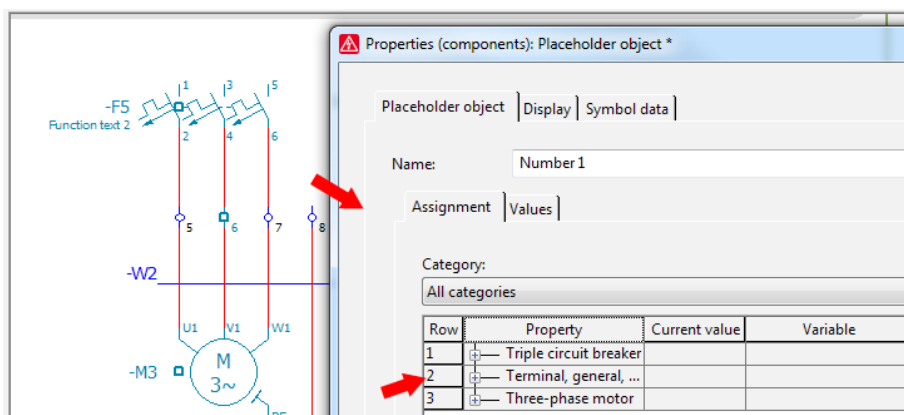


Fig. 5.214 Assigning individual objects to the placeholder object (option 1)

Option 1: You first select all objects and then use the **INSERT/PLACEHOLDER OBJECT** menu to insert the placeholder object. After you place the placeholder object, EPLAN opens the **PROPERTIES (COMPONENTS) PLACEHOLDER OBJECT** dialog, and the data can be edited. This approach is recommended, because it allows you to accurately select the objects that are to be applied to the placeholder object.

Option 2: You first insert the placeholder object in the page via the **INSERT/PLACEHOLDER OBJECT** menu. But before EPLAN places it, all objects must be selected with a window (pulled open with the mouse).

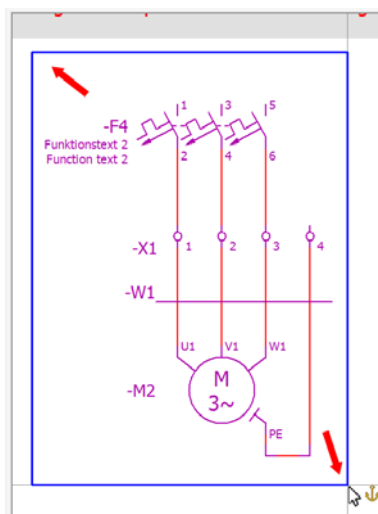


Fig. 5.215

Selecting objects with a window (option 2)



NOTE: Disadvantage of option 2: remote objects lying outside a window cannot be included in the value set.

Once you have chosen one of the methods, you can place the placeholder object. It is recommended that you place the placeholder object close to the macro. Simply use the left mouse button to move the placeholder object to the desired position and click to place it.

EPLAN immediately opens the **PROPERTIES (COMPONENTS) PLACEHOLDER OBJECT** dialog. Here you should enter a descriptive name for the placeholder object into the **Name** field.

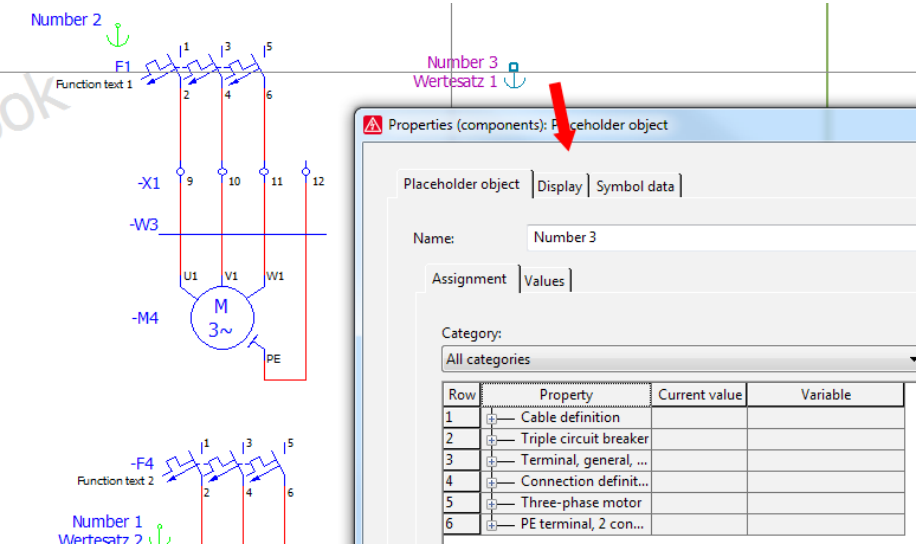


Fig. 5.216 Editing placeholder object

On the *Placeholder object* tab, you can now open the *Values* tab and define the variables that are to be later switched when the value set is selected. Right click in the free field and select the **NAME NEW VARIABLE** function.

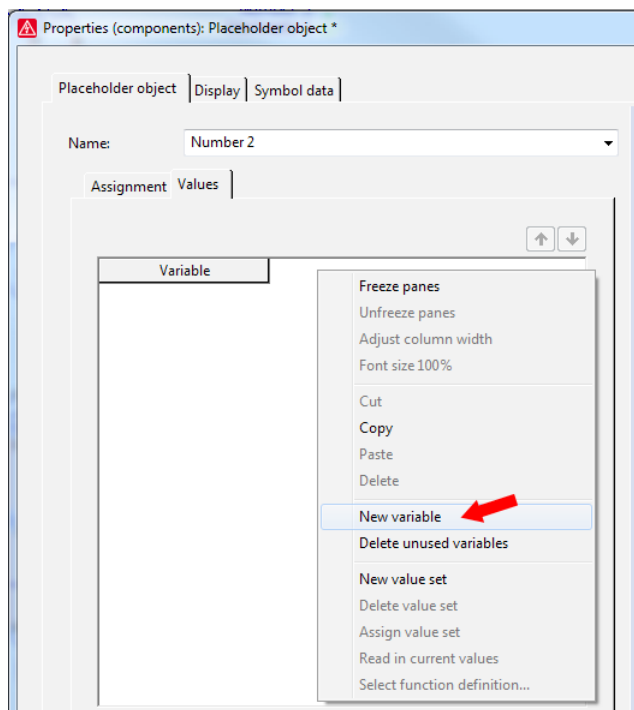


Fig. 5.217 Creating new variable

EPLAN opens the **NAME NEW VARIABLE** dialog. The *variable name* is now defined. When you click **OK**, the variable is applied in the *Values* tab.

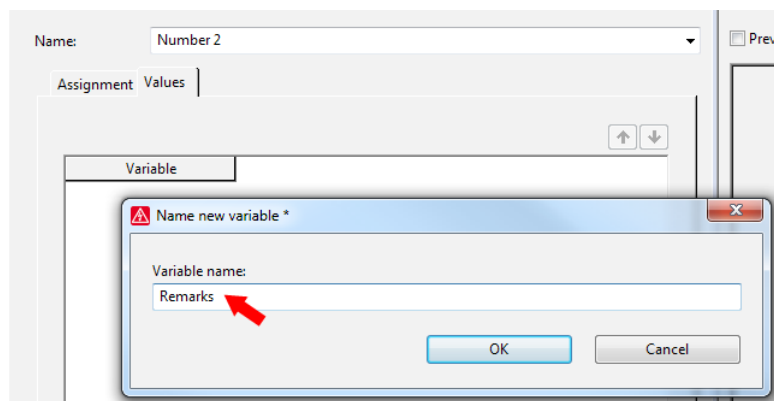
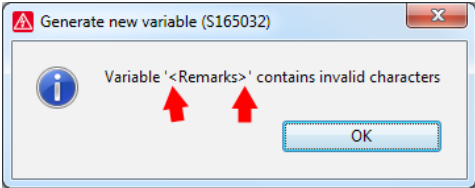


Fig. 5.218 Naming new variable

Fig. 5.219
Messages about characters not permitted (»<« or »>«)

If the variable name contains characters that are not permitted, such as angle brackets, EPLAN will display a message to this effect and will not apply the variable.



The value sets, which will be available for selection later on, must now be defined so that data is entered into the variables for the different value sets.

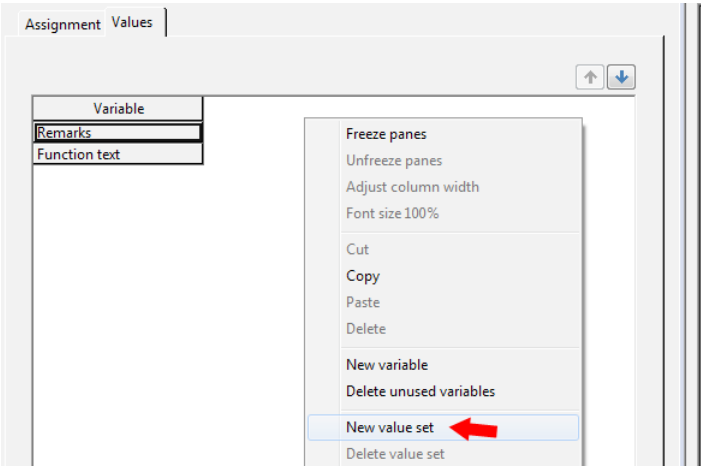


Fig. 5.220 Creating new value set

Click the right mouse button again, and select the **NEW VALUE SET** function. EPLAN inserts a new column next to the **Variable** column. Enter the desired designation of the value set into the header of the column. Proceed in the same way for the next new value set. You will now have several value sets at your disposal for the subsequent selection of “switchable” values.

Assignment Values		
Variable	Value set 1	Value set 2
Remarks	Remarks 1	Remarks 2
Function text	Function text 1	Function text 2

Fig. 5.221
Filling value sets with data

Data (the actual values) for the variables can now be entered below the value set name. To assign the variables (and their values) to the device properties, you switch to the *Assignment* tab, and select the property that is later to be switched.

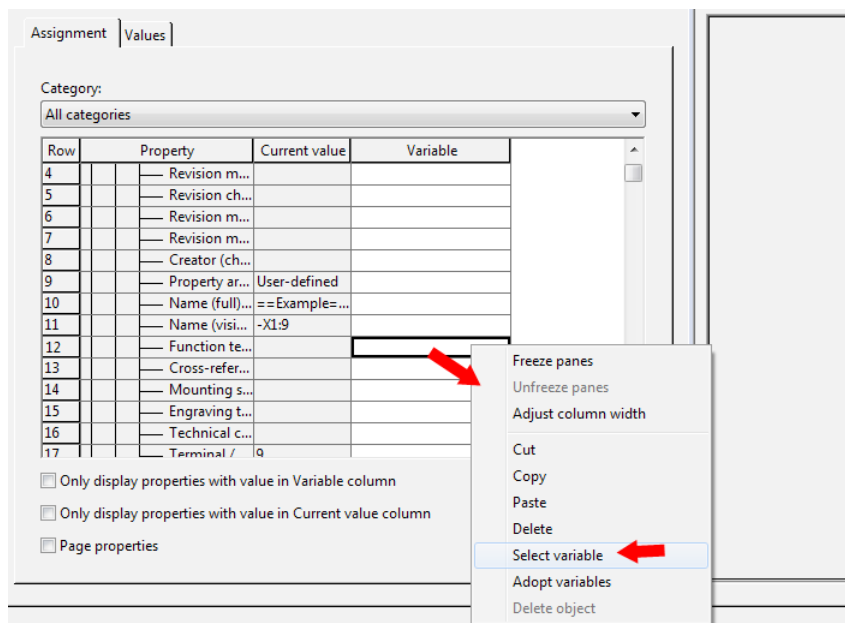


Fig. 5.222 Selecting variable

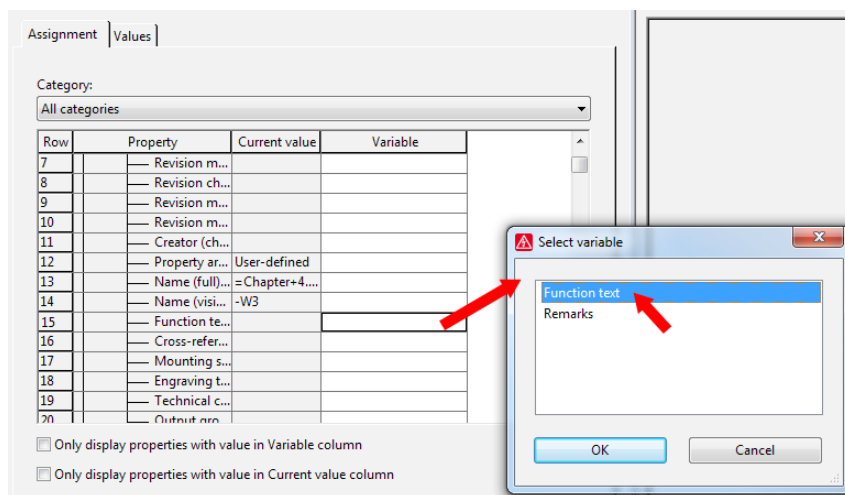


Fig. 5.223 Applying variable

Now you can select the variable from the **SELECT VARIABLE** dialog and apply it. EPLAN applies the variable and enters it correctly in the **Variable** column.



TIP: This approach is to be recommended, since variables can be simply applied from a dialog, and EPLAN enters these with the correct syntax. This excludes the possibility of incorrect entries, which is not always the case with manual entry.

Once the variable has been applied, EPLAN establishes the assignment between the variable value and the value set (name). The dialog can now be saved and closed with OK. To now switch between the individual value sets, you select the placeholder object and right click to select the **ASSIGN VALUE SET** command from the popup menu. EPLAN opens the **SELECT VALUE SET** dialog. Here you select the desired entry and apply it by clicking OK. EPLAN now switches over to the variables/values inserted into this value set. In the example, that could be the function text.

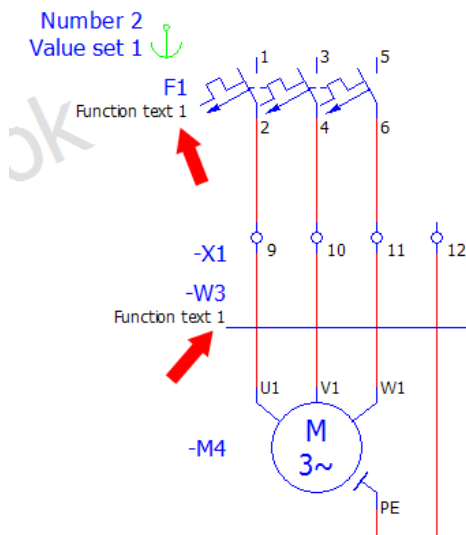


Fig. 5.224 Example value set 1

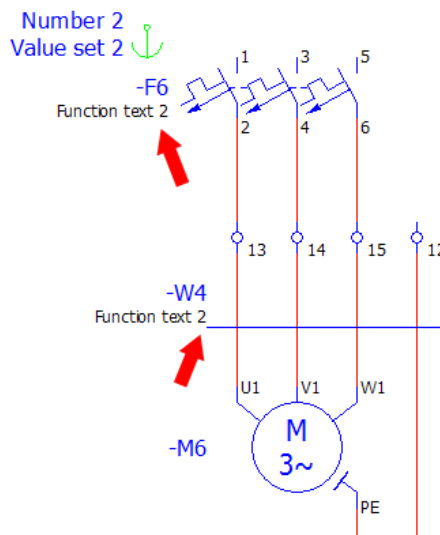


Fig. 5.225 Example value set 2

In conclusion, it can be said that this type of macro can be used in many different ways in projects. The example shown here was deliberately simple, merely pointing out the option of macros with value sets. The number of potential ideas is unlimited and possible sources of errors, e.g. incorrect part data or wrong cable cross-sections when changing motor power, are eliminated once and for all.

■ 5.9 Navigators – general functions

Most navigators and their popup menus contain functions of the same functional extent. Some of these functions are described one time below. Depending on the type of navigator, the view of certain devices may change (for example, the terminal strip navigator views terminals, while the cable navigator looks at the cable devices), but the functions of *Place* and *Assign* do the same things in all of these navigators. This is why these explanations have been limited, for the most part, to the device navigator.

5.9.1 Place from navigators

In the example in section 5.2.9, a device was inserted into the navigator. It was inserted in the navigator as an unplaced device. To put this device in the schematic (i.e. to place it), you select the **PLACE** function in the popup menu of the **device navigator**.

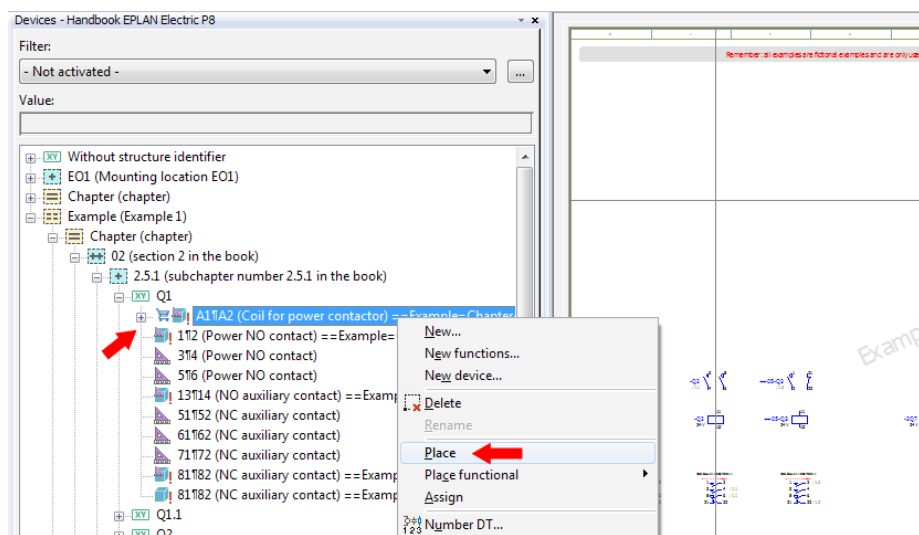


Fig. 5.226 Placing device

The device then “hangs” on the cursor and can be placed on the schematic page. It is also possible to drag more than one device onto the page. Multiple devices can be marked in the device navigator and placed on the page. EPLAN then places an item every time the mouse is clicked.

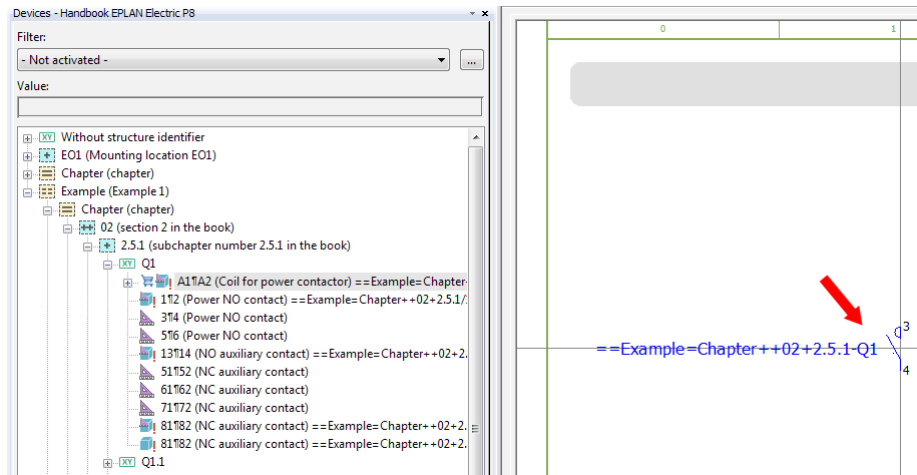


Fig. 5.227 Placed device

The major advantage of using the device navigator for placement is that EPLAN updates and enters all important information and data. This means that you won't have any incorrect DTs or missing technical characteristics on the device because EPLAN enters the correct (existing) information from the navigator.

EPLAN always places the device in accordance with the page type. For example, if the page is a *Schematic multi-line* page type, the *Schematic multi-line* representation type will be placed. If you do not want that, you must press the backspace key before placing the symbol on the page. EPLAN then opens the **PLACE DEVICE** dialog.

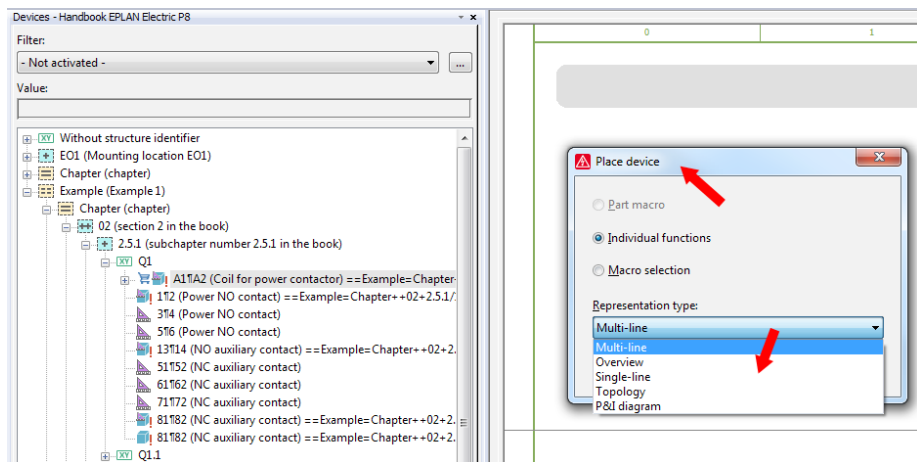


Fig. 5.228 Place device dialog

There are several options in this dialog:

Part macro: If the device has a part macro, you can select it here.

Individual functions: This option allows for all functions to be placed individually. Here, before placement (the **PLACE DEVICE** dialog is closed), use the **N** key to browse the functions. Thus, in the example, this is the circuit breaker, followed by the auxiliary NO contact, and finally the auxiliary NC contact.

Macro selection: Use this option to select a completely macro to be placed.

If only one of the options is selected and confirmed via **OK**, it is still possible to change the symbol before the actual placement. Again, the backspace key needs to be pressed. EPLAN then opens the **SYMBOL SELECTION** dialog.

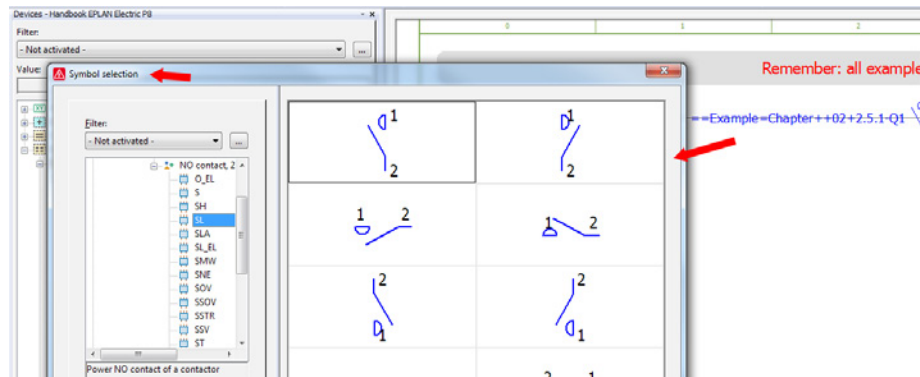


Fig. 5.229 Changing symbol before placement

If a symbol has been selected, the device can be placed fully, that is, all functions individually, for example. It is also possible here to rotate the symbol before placement by using the **TAB** key, that is, by changing the variant.

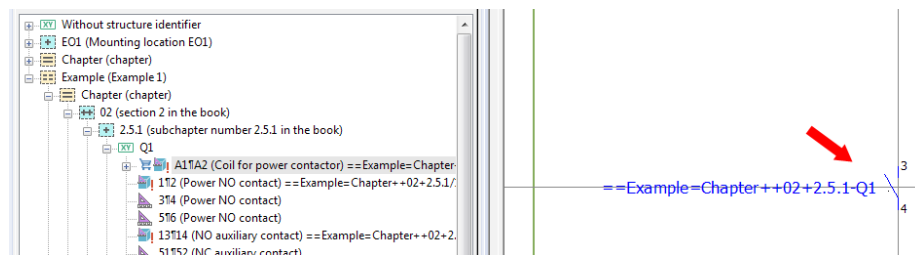


Fig. 5.230 Selecting variant before placement

5.9.2 Assign from navigators

A number of navigators have the **ASSIGN** function in their popup menu.

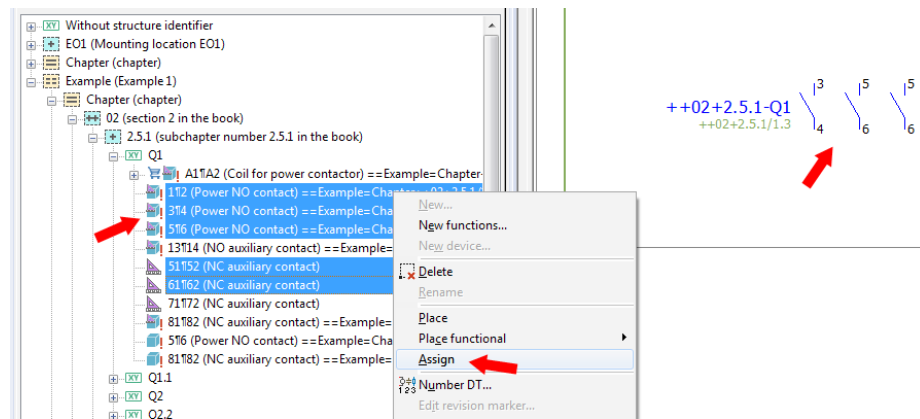


Fig. 5.231 The Assign menu item in the device navigator

The **ASSIGN** menu item allows the *properties of objects* selected in the navigator, e.g. a *device tag* or parts of a device such as a PLC connection point, to be assigned the properties of an existing object (e.g. a power contact).

It is also possible to select several objects in the navigator, execute the **ASSIGN** function, and then assign these properties to other objects by clicking them in the schematic.



You select a few devices in the **device navigator** and then execute the **ASSIGN** function. EPLAN now visibly “hangs” all of the properties for this device, such as for example the *device tag* or the *function definition*, on the cursor (the remaining unassigned device tags remain visible in the foreground).

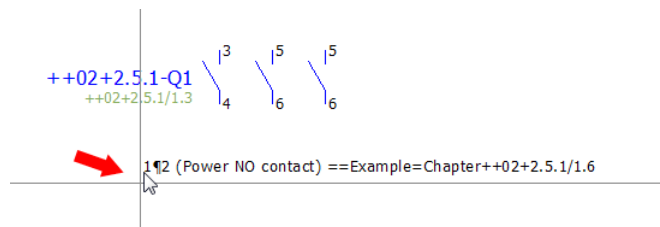


Fig. 5.232 The assigned properties “hang” on the cursor.

You then click the individual power contacts to be “equipped” in the schematic one after another, using the left mouse button.

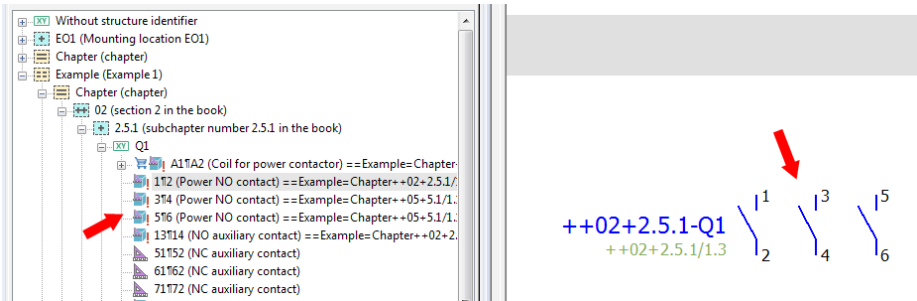


Fig. 5.233 Assigned device tags

EPLAN transfers one DT after another to the power contacts without requiring this to be manually added at each power contact. This is a very useful function that saves time and ensures the error-free transfer of information. It is also available in other navigators (in addition to the device navigator).

5.9.3 Filters

Filters are useful in displaying specific data in the navigators. There are really only two types of filters – filters without quick entry and those with quick entry.

5.9.3.1 Default filter

A default filter consists of one or several criteria that must be “fixed” in the filter.

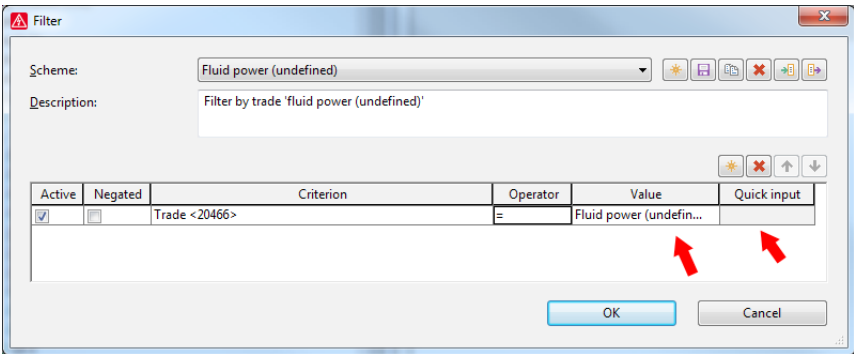


Fig. 5.234 Default filter, without quick entry

5.9.3.2 Quick entry filter

To simplify work, most navigators can also filter by a scheme according to different values. In other words, quick entry. This way, it is not necessary to create a filter for each case. Simply create a general filter with the option of activating quick entry. The actual filtering is then done via the selection field directly in the navigator.

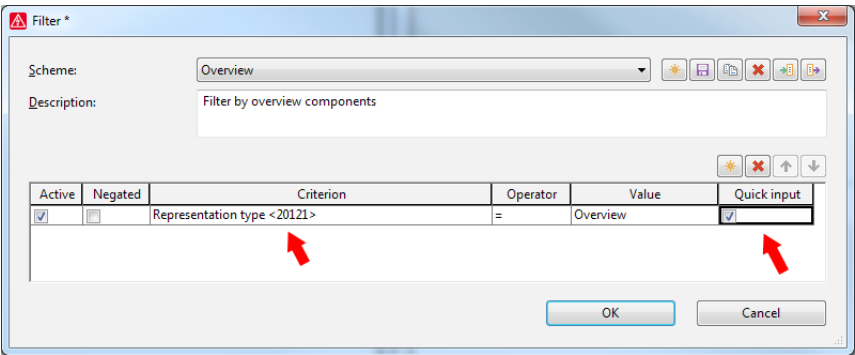


Fig. 5.235
Filter with quick entry
option

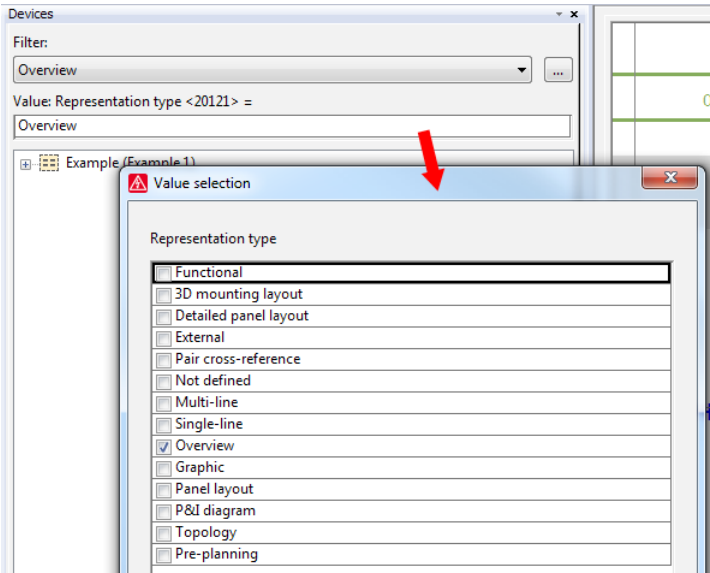


Fig. 5.236
Filter with quick entry

This type of filter, i. e. the ability to use quick input, eliminates the need to create individual schemes for each value.

In the **Device navigator** example shown, this would be filtering by representation type. Previously, you had to create a scheme for each of the representation types that you wanted to use for filtering. This resulted in a large number of possible, selectable schemes, which over time would mean that selecting a scheme would be confusing and cumbersome.

The ability to use the **QUICK ENTRY** function for properties reduces the number of different schemes that have to be created, so that is again clear and easy to select schemes.

 **NOTE:** Not all criteria are suitable for the quick entry option when filtering.

■ 5.10 Correction functions

EPLAN offers a number of different correction functions. In versions prior to EPLAN P8 V2.5, the correction functions were located in the menus of the respective navigators.

Now, these correction functions can be found in one place, in the **PROJECT/ORGANIZE/CORRECT** menu. In this menu, you can save and execute in your own filter schemes possible corrections for the areas *Terminals*, *Plugs*, *Cables*, and *Connections*.

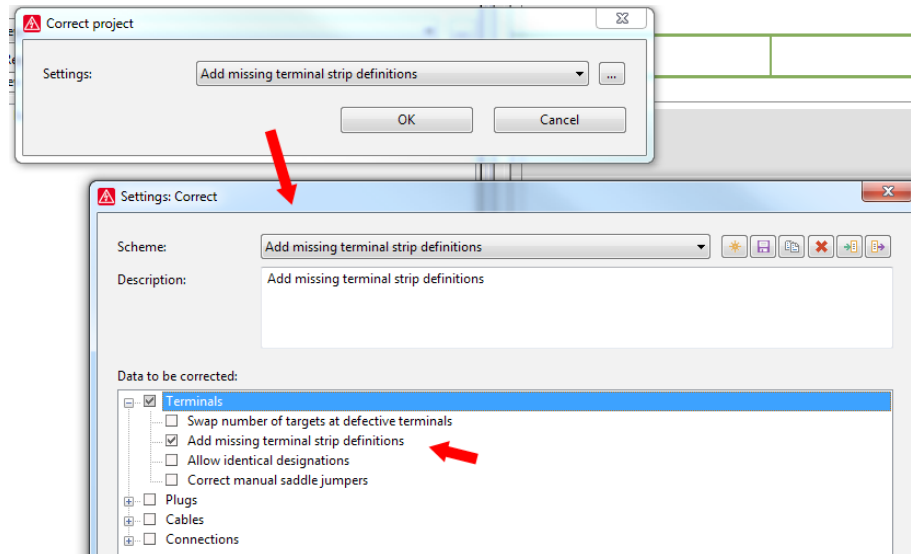


Fig. 5.237 Example of scheme for correcting terminals

6

Reports

This chapter deals with reports: Which ones are necessary in regular cases, what settings are needed for which reports, how do I make the most effective use of templates and reports, and what are the advantages for the user from using reports? Some of the possibilities offered by external editing and labeling are also touched upon using simple, short examples.

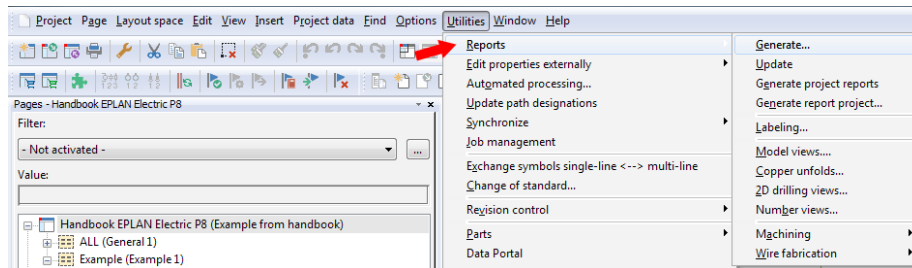


Fig. 6.1 Reports menu

The **UTILITIES/REPORTS** menu contains the functions for generating graphical reports such as terminal diagrams, parts lists and connection lists. The **UTILITIES/REPORTS/LABELING** menu item allows the user to export various reports such as cable overviews or summarized parts lists in (e.g.) Excel format.

EPLAN thus allows users to generate internal outputs (reports = graphical outputs for project documentation) and to transfer evaluated data into an external format (Excel, text format, etc.), for passing on the generated data from a schematic, for example.

Generation of graphical reports is only one option offered by EPLAN. Previously, you could externally edit the generated data via the *Labeling* function, but there was no way of synchronizing this with the EPLAN project, so an additional functional scope was added to EPLAN allowing properties to be exported, externally edited, and then re-imported back into EPLAN.

The **UTILITIES/EDIT PROPERTIES EXTERNALLY** menu item contains the functions allowing you to more conveniently edit a wide range of properties, in Excel for example,